

Finite Identities and Explicit Error Budgets in a Framework for Binary Goldbach

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Abstract

We consolidate a research corpus on binary Goldbach into a support-preserving framework with a documented verification boundary. Exact Fejér grid covariance and Ramanujan-proxy identities yield short-average variance bounds, including an even-support correction and a logarithmic refinement. Arithmetic reductions retain actual Vaughan coefficients, Möbius signs, coprimality masks, divisor caps and moving endpoints. The canonical Lean4 inventory contains 658 individually axiom-audited declarations; the last cycle contributes 19 finite conclusions. Written reductions to a reported primitive-character remainder input give a conditional universal reference gap $N - F_N \geq N/(256 \log N \log \log N)$ and a conditional effective exterior assembly for $\alpha = \lceil N^{1/4} \rceil$. The input's full direct remainder substitution remains outside the self-contained verification in this report. A separate classical sieve argument at $\alpha = \lceil N^{1/8} \rceil$ gives a negative composite sector and a one-sided bound for its rough complement, with an unevaluated onset. These profiles are not combined. We supply statements, parameter and dependency tables, explicit estimates, a formal appendix and reproduction material. The global signed remainder, a complete positive prime-pair margin and finite coverage below the analytic thresholds remain open. No proof of binary Goldbach or claim to overcome the parity barrier is made.

Keywords: binary Goldbach; Möbius correlation; Vaughan coefficients; explicit error budgets; sieve methods; Lean4.

Mathematics Subject Classification: 11P32.

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1 Main results, scope and conventions

Binary Goldbach asks whether every even integer $N \geq 4$ is a sum of two primes. This report closes a research phase of Project Horizon by separating exact finite identities, written analytic arguments, reproducible computations and remaining hypotheses. It concentrates on binary Goldbach. Neither the global signed estimate needed here nor an unconditional positive prime-pair margin has been proved.

The mathematical objects are the actual Möbius function μ , von Mangoldt function Λ and Euler totient φ . We write $u = \log N$, $L = 1 + u$ and $\ell = \log u$ when $u > 1$. All logarithms are natural. Indicator functions are denoted by $\mathbf{1}$. Strict inequalities in integer selectors are literal; replacing an integer floor by a real endpoint can change a coefficient. The Goldbach singular factor is

$$\mathfrak{S}(N) = 2C_2 \prod_{\substack{p|N \\ p>2}} \frac{p-1}{p-2}, \quad C_2 = \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right) \quad (1)$$

for even N . The same expression defines $\mathfrak{S}(K)$ for an even positive mask K . A reference term involving $\mathfrak{S}(N)$ is distinct from an actual count of prime pairs.

1.1 Four contributions and their scope

The report makes four concrete contributions within a partial research framework.

1. It gives explicit finite Fejér covariance and Ramanujan-proxy identities, with the even-support main term corrected and the proxy energy refinement stated with its exact constants. These use classical harmonic tools and do not estimate the true prime-pair remainder.
2. It assembles support-preserving Vaughan, character and divisor identities with a terminal error ledger. The finite Lean statements retain actual coefficients and literal boundaries; they do not supply the missing analytic cancellation.
3. It exposes the dependency chain for explicit exterior allowances in the quarter profile, conditionally on the stated primitive-character input, and proves a separate eventual signed sector estimate in the eighth profile using classical sieve arguments. The two profiles remain distinct.
4. It supplies a versioned reproduction record connecting formal statements, retained compiler evidence, rational checks and written deductions. Specific counterexamples delimit particular naive envelopes; they do not establish the failure of every possible analytic method.

These contributions concern precise formulation, explicit calculations and finite formalization. The declaration count is an inventory measure, and priority or novelty beyond these descriptions is not claimed.

1.2 What this report establishes and what it does not

The earlier geometric stage gives exact Fejér grid defects and gcd covariance, actual Ramanujan-proxy variance bounds and a correction of the proxy's even-support main term. The later arithmetic stage retains the actual Vaughan coefficients, coprimality masks, character corrections and moving support. Its finite conclusions have been formalized in Lean4; its analytic distribution, zero-free-region and sieve inputs have not.

The most recent written arguments reduce a universal reference gap and a scoped effective exterior assembly to explicit reported analytic inputs in one parameter profile. A signed negative composite sector has a separate classical proof in a different profile. The direct primitive-character remainder substitution is documented in the retained sources, but has not been completely reproduced in this manuscript; the consequences depending on that input are stated conditionally. The finite ingredients include literal divisor completion, the exact small-divisor credit and cutoff invariance. They do not by themselves prove the mass of a prime sector or the smallness of the complementary signed sum.

Result family	Retained conclusion	Evidence and boundary
Fejér and Ramanujan geometry	Exact covariance; short-centre proxy bounds; corrected even main	Written proofs and finite formal ingredients; proxy only
Finite arithmetic infrastructure	658 individually audited declarations across 71 canonical sources; 19 latest public conclusions	Retained compiler/axiom evidence; imports and copies not recounted
Universal reference	Conditional bound $N - F_N \geq N/(256u\ell)$ for $u \geq 8 \cdot 10^6$	Reduction to the explicitly reported primitive-character input
Quarter-profile exterior	Conditional assembly $\varepsilon_{\text{tr}} < (8 \cdot 10^{-6} + 2 \cdot 10^{-12})N/(u\ell)$ for $u \geq 10^{24}$	Reported primitive input and completion bridges; no Lean analytic certificate
Eighth-profile sector	$F_C + F_{\text{rough}} < -N/3000$ eventually	Classical sieve deduction; onset unevaluated
Global signed remainder	A sufficient target is $D_N \leq N/(256u\ell)$ in the quarter profile	Open; covered bridge is a separate premise

Table 1: Evidence classes. “Audited” refers to the retained source, compilation and axiom records; it does not mean that the analytic estimates have been checked by Lean.

1.3 Relation to existing work

The circle-method reference in (1) is classical [1]. The prime/Möbius transfer belongs to the same general problem as the twisted distribution hypotheses studied by Huang–Li [4]. The proxy estimates use ordinary Ramanujan algebra and the large sieve [5, 6]; the sector counts use classical sieve distribution, rather than a new theorem overcoming the parity barrier. Recent work on the Goldbach exceptional set is surveyed by Bhowmik–Grimmelt [2]. This report claims no improved exceptional-set exponent and no new formula generating primes.

The defensible contribution is a source-bound formal and analytic research synthesis: explicit support-preserving identities, particular constants and error components, and a reproducible record of the residual arithmetic obstruction. Priority for a standalone analytic result or a broader publication claim has not been established. Finite computational verification, including the published range up to $4 \cdot 10^{18}$ [3], cannot bridge the interval below $\exp(10^{24})$ if a future argument only works above that onset.

Profile	Parameters	Admissibility and status
Earlier polylogarithmic Vaughan	Fixed $U, V \leq L^{B_0}$; $Q \leq C_0 N^{1-\theta}$	Written Type-I removal; constants not all numerically evaluated
Effective terminal quarter	$U = V = 1$, $\alpha = \lceil N^{1/4} \rceil$, $Q = \lfloor (N-1)/\alpha \rfloor$	Common onset $u \geq 10^{24}$ for the final exterior assembly
Qualitative signed eighth	$U = V = 1$, $\alpha = \lceil N^{1/8} \rceil$, $Q = \lfloor (N-1)/\alpha \rfloor$	Classical eventual sector estimates; no evaluated common exterior budget
Auxiliary square/conductor cuts	$B = \lceil L^{20} \rceil$, $R_c = \lfloor L^{16} \rfloor$	Used with their stated profile; not independent free parameters

Table 2: Profiles that must remain separate. The onset $u \geq 10^{24}$ is not assigned to the eighth profile.

1.4 Reading the proof status

An exact finite theorem may be fully kernel checked while the subsequent analytic consequence remains a written argument using external theorems. Rational scripts check algebraic comparisons, exponents and finite configurations; they do not verify a region free of zeros or an asymptotic sieve estimate. Repeated archive hashes certify provenance. Reviews by separate AI agents are internal checks, and are not external peer review. The electronic registry records the distinction for each formal declaration and excludes the weak historical interfaces described in Section 14.1.

2 Ordered pairs and the CS21 reference audit

Let

$$G(N) = \#\{p \leq N/2 : p, N-p \text{ prime}\}, \quad R_2(N) = \#\{(p, q) : p+q = N, p, q \text{ prime}\}.$$

Then $R_2(N) = 2G(N) - \mathbf{1}_{N/2 \text{ prime}}$. The baseline

$$B_0(N) = C_2 \prod_{\substack{p|N \\ p>2}} \frac{p-1}{p-2} \frac{N}{\log^2 N}$$

is an unordered baseline. The original CS21 residual used $G(N) - 2B_0(N)$ [18]; the factor2 belongs to the ordered count. A more accurate classical unordered reference is

$$B_{\text{int}}(N) = C_2 \prod_{\substack{p|N \\ p>2}} \frac{p-1}{p-2} \int_2^{N-2} \frac{dt}{\log t \log(N-t)}.$$

Correcting a baseline changes the residual being predicted, not the prime data.

The reproduction uses $M = 997$, weights $p^{-1/2}$, 23 nonzero modes in $\{0, 1, 2, 3\} \times \{0, 1, 2, 3, 4, 5\}$, training-only standardization and ridge parameter1 with an unpenalized intercept. The five $k = 0$ features do not depend on M and account for the original test signal. On this window the signal disappears after the integral reference. No new permutation-significance test or universal absence of subleading arithmetic structure is claimed. Further windows and perturbations are preserved in the companion CSV.

Residual reference	Predictor	Test correlation	Test R^2
$2B_0$ (original)	23 torus modes	0.178892	0.031662
$2B_0$ (original)	Five $k = 0$ modes	0.178941	0.031773
B_0 (normalized)	23 torus modes	0.154728	0.021582
B_{int}	23 torus modes	-0.013187	-0.030667

Table 3: Retained CS21 reanalysis on the second half of [120000, 240000]. Parameters were fit on the first half. These are floating-point diagnostics, not formal asymptotic theorems.

Two mathematical checks explain why this distinction matters. First, fixed-modulus features of the form

$$\sum_{p \leq N} p^{-1/2} \cos \left(2\pi \left(\frac{kp}{M} + \frac{j \log p}{\log X} \right) \right)$$

have leading term $(c_M(k)/\varphi(M)) 2\sqrt{N}/\log N$, uniformly for N/X in a fixed compact subset of $(0, \infty)$, for fixed M, k, j . The PNT in fixed residue classes and Abel summation prove this: on $[\epsilon X, bX]$ the logarithmic phase tends uniformly to 1, and the smaller primes cost $O(\sqrt{\epsilon X}/\log X)$. Taking $X \rightarrow \infty$ then $\epsilon \rightarrow 0$ gives the claim. Standardization can amplify lower terms, so this is not a theorem about the fitted ridge model.

Second, the classical PNT and summation by parts yield, for each fixed $A > 0$,

$$\sum_{\substack{4 \leq N \leq X \\ N \text{ even}}} (G(N) - B_{\text{int}}(N)) = O_A(X^2 \log^{-A} X). \quad (2)$$

For completeness, expand $f(m) = \prod_{p|m, p>2} (1 + 1/(p-2))$ as a divisor sum with coefficient $a(d) = \mu(d)^2 \mathbf{1}_{2 \nmid d} \prod_{p|d} (p-2)^{-1}$. Its mean is $Y/C_2 + O_\epsilon(Y^\epsilon)$ because $\sum a(d)/d = 1/C_2$ and $\sum a(d)d^{-\epsilon} < \infty$. Twice replacing the prime measure by $dt/\log t$ gives the triangular double integral with an arbitrary logarithmic saving. Removing the prime2 pairs and the diagonal costs $O(\pi(X))$. Abel summation of the mean of f matches the integral reference, proving (2). This is signed cumulative agreement; it gives neither a bound at each even N nor positivity at that N .

3 Exact grid covariance and Ramanujan proxies

These results concern a geometric or Ramanujan proxy. They preserve useful arithmetic coefficients, but do not replace the convolution of two actual prime sequences.

3.1 The exact Fejér defect

For integers $H, d \geq 1$ and $s \in \mathbb{Z}$, put

$$B_{H,s}(n) = \left(1 - \frac{|n-s|}{H} \right)_+, \quad \Delta_{H,d}(s) = \sum_{d|n} B_{H,s}(n) - \frac{H}{d}.$$

The sum runs over all integers; it is finite because of the triangular support. Write $r = H \bmod d$, $v = s \bmod d$ with $0 \leq r, v < d$.

Theorem 3.1 (Exact defect and covariance). *One has*

$$\Delta_{H,d}(s) = \frac{(r-v)_+ + (r+v-d)_+ - r^2/d}{H}, \quad |\Delta_{H,d}(s)| \leq \frac{d}{4H}. \quad (3)$$

In particular it vanishes if $d \mid H$. Its mean over $s \bmod d$ is zero. If $g = \gcd(d, e)$ and $a_g = \min(H \bmod g, g - H \bmod g)$, then the covariance over a common period is

$$\mathbb{E}_s[\Delta_{H,d}(s)\Delta_{H,e}(s)] = \frac{g(2a_g^3 + a_g)/3 - a_g^4}{deH^2}. \quad (4)$$

Proof. The triangle is the autocorrelation of $\mathbf{1}_{\{0, \dots, H-1\}}$ divided by H . The residue counts of this interval are $t + \mathbf{1}_{\{0, \dots, r-1\}}$, where $H = td + r$. Its cyclic intersection at shift v is $(r-v)_+ + (r+v-d)_+$. Subtracting H/d gives (3). The largest positive deviation is $r - r^2/d \leq d/4$ before division by H ; complementing the interval gives the same bound on the negative deviation.

For $r \leq d/2$, the cyclic intersection has value r once, each value $1, \dots, r-1$ twice, and zero elsewhere. Its sums are r^2 and $(2r^3 + r)/3$. Complementing handles $r > d/2$, so

$$\text{Var}(\Delta_{H,d}) = \frac{1}{H^2} \left(\frac{2a_d^3 + a_d}{3d} - \frac{a_d^4}{d^2} \right).$$

Let $e(t) = \exp(2\pi it)$ and $\hat{B}_H(t) = H^{-1} |\sum_{j=0}^{H-1} e(jt)|^2$. Finite Fourier orthogonality gives

$$\Delta_{H,d}(s) = \frac{1}{d} \sum_{j=1}^{d-1} \hat{B}_H(j/d) e(js/d).$$

The covariance retains only the common opposite frequencies j/g , whence it equals $g^2/(de)$ times the variance with modulus g . This proves (4). $\square$

For arbitrary finite complex coefficients ω_d , the exact energy is therefore

$$\mathbb{E}_s \left| \sum_d \omega_d \Delta_{H,d}(s) \right|^2 = \sum_{d,e} \omega_d \bar{\omega}_e \frac{g(2a_g^3 + a_g)/3 - a_g^4}{deH^2}.$$

This retains covariance instead of separately paying every entry. Markov's inequality bounds the proportion of large defects. It gives no estimate at a prescribed centre, and a common period may vastly exceed the interval of interest.

3.2 Actual proxy coefficients and short averages

Let $c_q(n) = \sum_{a \bmod q, (a,q)=1} e(an/q)$ be the Ramanujan sum and define

$$A_Q(n) = \sum_{q \leq Q} \frac{\mu(q)}{\varphi(q)} c_q(n), \quad G_Q = \sum_{q \leq Q} \frac{\mu(q)^2}{\varphi(q)}, \quad \mathcal{P}_Q = \prod_{p \leq Q} \left(1 + \frac{9p}{(p-1)^2} \right).$$

The exact expansion $A_Q(n)^2 = \sum_g S_g c_g(n)$ is supported on squarefree $g \leq Q^2$, with primes at most Q . For squarefree g , set

$$\xi(t) = \prod_{p|t} \frac{p-2}{p-1}, \quad \Gamma_Q(t, v, w; g) = \sum_{\substack{j \leq Q/(t \max(v, w)) \\ (j, g)=1}} \frac{\mu(j)^2}{\varphi(j)}.$$

Then

$$S_g = \frac{\mu(g)}{\varphi(g)} \sum_{t, v, w=g} \mu(t) \xi(t) \Gamma_Q(t, v, w; g), \quad S_1 = G_Q, \quad |S_g| \leq \frac{G_Q 3^{\omega(g)}}{\varphi(g)}. \quad (5)$$

To verify this, use $c_p^2 = (p-2)c_p + (p-1)$. In a pair $q = t j v$, $r = t j w$, common primes retained in the Ramanujan term belong to t , common primes replaced by the constant belong to j , and

primes occurring in just one factor belong to v, w . These four factors are pairwise coprime and squarefree. Their actual cutoffs give the displayed Γ_Q ; there are $3^{\omega(g)}$ choices of t, v, w . The signs are present in the identity and only afterwards bounded absolutely.

Put $K_H(g) = \sum_{1 \leq a < g, (a, g) = 1} |\widehat{B}_H(a/g)|^2$. Parseval on a residue period gives $0 \leq K_H(g) \leq gH$: reduce the geometric sum to length $a_g \leq H$, and compare its triangular square norm with that of a nonoverlapping triangle modulo g . The centred proxy defect

$$\mathcal{E}_{H,Q}(s) = \sum_n B_{H,s}(n) A_Q(n)^2 - HG_Q$$

has complete-period energy $\sum_{g>1} S_g^2 K_H(g) \leq HG_Q^2 \mathcal{P}_Q$. Distinct reduced frequencies have denominators at most Q^2 and separation at least Q^{-4} . The dual large sieve [5] consequently yields

$$\frac{1}{T} \sum_{s=s_0+1}^{s_0+T} |\mathcal{E}_{H,Q}(s)|^2 \leq \left(1 + \frac{Q^4}{T}\right) HG_Q^2 \mathcal{P}_Q \ll H \left(1 + \frac{Q^4}{T}\right) \log^{11}(2Q). \quad (6)$$

The last estimate uses only classical harmonic prime bounds. It assumes no cancellation in a shifted Möbius correlation.

3.3 The parity correction and the later coefficient-energy refinement

On even integers the proxy compresses exactly:

$$A_Q(2m) = V_Q(m) := \sum_{\substack{Q/2 < q \leq Q \\ q \text{ odd}}} \frac{\mu(q)}{\varphi(q)} c_q(m). \quad (7)$$

Pair q with $2q$ for odd $q \leq Q/2$ and use $c_{2q}(2m) = c_q(m)$ to see the cancellation. For $H = 2h, s = 2t$, the even-support sum becomes $\sum_m B_{h,t}(m) V_Q(m)^2$. Its main term is

$$hG_Q^{\text{term}}, \quad G_Q^{\text{term}} = \sum_{\substack{Q/2 < q \leq Q \\ q \text{ odd}}} \frac{\mu(q)^2}{\varphi(q)} \longrightarrow \frac{\log 2}{2}.$$

For $Q = 6$ the correct main is $H/8$, whereas incorrectly using $(H/2)G_Q$ gives $13H/8$.

The elementary dyadic bound $\sum_{x < n \leq 2x} 1/\varphi(n) < 2$ follows from $n/\varphi(n) = \sum_{d|n} \mu(d)^2/\varphi(d)$, the harmonic dyadic sum at most 1, and $\prod_p (1 + 1/[p(p-1)]) < 2$. Applying it to (5) bounds the terminal coefficients by $2 \cdot 3^{\omega(g)}/\varphi(g)$ and replaces the exponent 11 in (6) by 9. The limit of G_Q^{term} follows by convolving the odd harmonic sequence with the absolutely summable local correction $b(p) = 1/[p(p-1)]$, $b(p^2) = -1/[p(p-1)]$.

Define the even-support defect explicitly by

$$\mathcal{E}_{h,Q}^{\text{even}}(t) = \sum_m B_{h,t}(m) V_Q(m)^2 - hG_Q^{\text{term}}.$$

The subsequent finite coefficient audit improves this further. If $V_Q^2 = \sum_g R_g c_g$, define

$$\mathcal{K}_Q = \prod_{\substack{p \leq Q \\ p \text{ odd}}} \left(1 + \frac{2p-1}{p(p-1)^2}\right), \quad H_Q^{\text{harm}} = \sum_{n \leq Q} \frac{1}{n},$$

and

$$\begin{aligned} P(x) = 1 &+ \frac{31x}{12} + \frac{17x^2}{12} + \frac{739x^3}{2592} + \frac{199x^4}{7776} \\ &+ \frac{17x^5}{15552} + \frac{x^6}{46656} + \frac{x^7}{6531840}. \end{aligned}$$

The retained arithmetic coefficient theorem is

$$\sum_g gR_g^2 \leq 8\mathcal{K}_Q^{11} \min\{(H_Q^{\text{harm}})^7, P(2\log Q)\}. \quad (8)$$

Its finite reconstruction assigns each prime in a quadruple of squarefree moduli to one of eleven incidence patterns. Four anchor fibres have total mass at most $8\mathcal{K}_Q^4$; the seven remaining variables have full harmonic masses or a constrained simplex mass. Their degrees are $(d_1, \dots, d_7) = (2, 2, 3, 3, 3, 3, 4)$ and their actual product constraint is $\prod_j D_j^{d_j} \leq Q^4$. The positive odd harmonic simplex bound expands to

$$P(t) = \sum_{I \subseteq \{1, \dots, 7\}} \frac{t^{|I|}}{|I|! \prod_{j \in I} d_j},$$

evaluated at $t = 2\log Q$; the four terminal anchor diameters are $(4, 4, 2, 4)$, which give the factor 8. The source-bound finite proof, including the coefficient identity and that polynomial identification, is in the canonical phase 4 modules and the electronic statement registry. The full harmonic sum used above is a harmless larger envelope for the formal odd harmonic sum. The resulting variance transfer is still a written Fourier/large-sieve argument; it is not identified with an actual prime-pair variance theorem in Lean. Since $\mathcal{K}_Q$ is bounded by an absolutely convergent Euler product, (8) gives

$$\frac{1}{T} \sum_{t=t_0+1}^{t_0+T} |\mathcal{E}_{h,Q}^{\text{even}}(t)|^2 \ll h(1 + Q^4/T) \log^7(2Q). \quad (9)$$

For $h \asymp X^\theta$, $Q \asymp X^\kappa$, $T \asymp X$ and $4\kappa < 1 + \theta$, Chebyshev gives a relative defect tending to zero for almost all centres, for any power saving $X^{-\delta}$ with $2\delta < \min(\theta, 1 + \theta - 4\kappa)$. Neither the Q^4/T loss nor the exceptional centres are removed. The result is an explicit internal proxy improvement using classical tools, not a resolution of the parity obstruction.

4 Finite additive certificates and limits of linear information

Theorem 4.1 (A phase-preserving finite certificate). *For real f on $\mathbb{Z}/M\mathbb{Z}$, put $E = \sum_x f(x)^2$ and $C_k(N) = \sum_x f(x) \cos(2\pi k(x - N/2)/M)$. For any distinct residues K ,*

$$\sum_x f(x)f(N-x) \geq \frac{2}{M} \sum_{k \in K} C_k(N)^2 - E.$$

If f is the indicator of primes at most $N-2$ and $M > 2(N-2)$, a positive right side proves an actual prime representation of N .

Proof. Set $z_k = e(-kN/(2M)) \sum_x f(x)e(kx/M)$. Fourier convolution and the identity $\Re z^2 = 2(\Re z)^2 - |z|^2$, together with Parseval, give the equality with all M modes. Dropping nonnegative real squares gives the inequality. The stated modulus prevents sums of two support elements from wrapping around. $\square$

At $N = 100$, $M = 201$, $E = 25$, the 19 modes

$$\{0, \pm 1, \pm 3, \pm 15, \pm 21, \pm 40, \pm 60, \pm 67, \pm 99, \pm 100\}$$

have a rigorously evaluated lower side exceeding 0.417911728893458011239466727237. Rational intervals for π and cosine, with explicit Taylor remainders, certify the sign. The mode selection used the full spectrum, so no efficient uniform mode-selection algorithm follows. Larger FFT

examples are diagnostics. The later finite formalization and the rational interval computation have separate evidence records.

A different limitation is elementary. For every fixed B , one can remove only $O(x/\log^2 x)$ primes below x , preserve all primes below B and every fixed reduced-class PNT, yet create infinitely many holes in the sumset of the remaining primes. Indeed take $X_j = 4^j X_0 > 2B$. By averaging pairs over even targets in $[X_j, 2X_j]$, choose N_j with $O(X_j/\log^2 X_j)$ unordered representations. Remove the larger member of each pair. Each removed member belongs to $[X_j/2, 2X_j]$ and determines its unique partner, so summing the geometric scales gives the claimed removal bound at every x . The remaining set has the same fixed-class densities, and its normalized Mangoldt Fourier transform differs from that of the primes by at most $O(1/\log x)$, uniformly in phase. Every N_j is nevertheless missing from its binary sumset.

This construction is not a counterexample to Goldbach for actual primes. It shows why qualitative linear density or Fourier proximity at that rate does not supply the desired pointwise binary conclusion. Likewise an absolutely summable linear zero expansion $Z(x) = \sum_\rho c_\rho x^\rho$ with $\Re \rho < 1$ satisfies $Z(x) = o(x)$ by finite-core/tail splitting; that qualitative fact alone does not prove a bound on the additive quadratic prime correlation.

5 From finite Vaughan fibres to the signed arithmetic remainder

The standard actual coefficient

$$\beta_V(b) = \sum_{\substack{d|b \\ d>V}} \Lambda(d)$$

obeys $\sum_{ab=n} \mu(a)\beta_V(b) = \Lambda(n)\mathbf{1}_{n>V}$. This is Dirichlet inversion of the divisor identity for $\log n$, rather than a new prime-generation formula. Splitting the a fibre at U gives the actual short component and the Type-II component used later. A completed fibre cancels composites, but an incomplete dyadic rectangle need not.

Expanding the real coefficient energy produces the four signs

$$\mu(a)\mu(c)\mu(N-ab)\mu(N-cb),$$

with the actual $\beta_V(b)$ weights, support and centring. Their smallness does not follow from orthogonality. Retaining $\mu(a)$ in a signed correlation is more informative than absorbing it in a generic L^2 norm, but its quantitative cancellation remains an arithmetic input. On the coprime arithmetic line, $a = a_0 + kt$, $r = r_0 - bt$, preserving the linear relation $ba + kr = N$. The CRT divisor modulus in the separate two-divisor expansion remains ar . Replacing the retained coefficients by a bare $\mu(a_0 + kt)\mu(r_0 - bt)$ correlation requires a proved transfer.

The exact completed character decomposition $R_k = B_k + C_k + E_k$ keeps nonunits, the mask $(n, Nk) = 1$, and the unit correction. Character orthogonality diagonalizes residues; it does not diagonalize the arithmetic four-sign correlation. For the actual squarefree primitive trace on unit- n , unit- N support,

$$P_q(N-n) = \prod_{p|q} ((p-1)\mathbf{1}_{p|N-n} - 1),$$

one has $P_q(m) = \mu(q)$ if $(q, m) = 1$. Thus $\mu(q)P_q(m)/\varphi(q) = 1/\varphi(q)$ there: the two displayed factors are not independent sign oscillators. For a unit t , the map $x \mapsto t/x$ on the unit group in the relevant finite Kloosterman geometry is a permutation and preserves energy. An entrywise bound cannot by itself contract arbitrary coefficient directions.

The four energy sectors have different diagonals. If $\mathcal{E} = \sum_b w_b (\sum_{a,k} x_{a,k,b})^2$, let D_{00} keep $a = c, k = k'$, D_a keep $a = c$, and D_k keep $k = k'$. Removing nonzero shifts $c - a$ subtracts D_a , not just D_{00} . A multiplicity estimate $\mathcal{E} \leq MD_a$ is valid but too costly in the parameter range considered. Local multiplicity improves constants; large Gram eigenvalues only exclude particular uniform operator bounds and do not determine the projection of the actual Möbius direction.

The later HH expansion keeps coefficients $H_y(a)H_y(r)$ and the actual CRT modulus ar . The sign conductor T in a grouping $\sum_T \mu(T)A_T$ is not automatically the CRT modulus. A discrepancy $C_{a,r} - M_{a,r}$ can be signed, so positive masses must be defined before subtracting the model. The Rosser remainder uses $\Phi(n, p(e))$, where $\Phi(n, z) = \mathbf{1}_{(n, \prod_{p < z} p) = 1}$; it must not exclude the prime $p(e)$ itself [16]. On a fixed short cofactor sector $\zeta \leq L^B$ with fixed B , the written absolute amplitude improves to $NL^{3+o(1)}$. Setting another short rough divisor to 1 only changes two factors of type $L^{o(1)}$ and supplies no second full logarithmic saving. Neither estimate is the necessary signed global budget.

These reductions explain the final choice of exact scalar error accounting. No enumeration of finite identities, nor a sharper proxy amplitude, replaces the missing bound for the aggregate on its complete arithmetic support.

6 The literal terminal scalar and its transfer identities

All logarithms in this section are natural. We use the actual arithmetic functions μ , Λ and φ . Let $N \geq 2$ be an integer, put $u = \log N$, and, when $u > 1$, put $\ell = \log u$. For even K define

$$C_2 = \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right), \quad \mathfrak{S}(K) = 2C_2 \prod_{\substack{p|K \\ p>2}} \frac{p-1}{p-2}. \quad (10)$$

The infinite product in this definition converges. The scalar

$$F_N = \sum_{1 \leq n < N} \Lambda(n) \mu(N-n) \quad (11)$$

is a signed prime–Möbius correlation. It is not a prime-pair count.

Let α, Q be positive integers, write $m = N - n$, and retain the literal kernels

$$D_Q(m) = \sum_{\substack{k|m \\ k \leq Q \\ \alpha k < m}} \mu(k) \log \frac{k}{m}, \quad (12)$$

$$W_Q(n, m) = \sum_{\substack{1 \leq k \leq Q \\ (k, nN)=1 \\ \alpha k < m}} \frac{\mu(k)}{\varphi(k)} \log \frac{k}{m}. \quad (13)$$

Inactive and empty sums are zero. In particular the strict inequality $\alpha k < m$ is part of the definition; it is not replaced by a real endpoint or an unproved completion. If

$$R = \min \left(Q, \left\lfloor \frac{m-1}{\alpha} \right\rfloor \right),$$

then the second kernel is the prefix through this exact integer R . For $V \geq 1$, set

$$\rho(n) = \mathbf{1}_{\{n > V\}} \mathbf{1}_{\{(n, N)=1\}}.$$

The complete fixed scalar is

$$\mathcal{F}_{\text{fixed}} = \sum_{1 \leq n < N} \rho(n) \Lambda(n) \mu(m) (W_Q(n, m) - D_Q(m)). \quad (14)$$

Every unit, divisor and moving-boundary restriction in this expression remains in force below.

Lemma 6.1 (The omitted-divisor coefficient). *For $m \geq 1$, define*

$$J_{Q,\alpha}(m) = - \sum_{\substack{d|m \\ d \leq \alpha \text{ or } m > Qd}} \mu(d) \log d.$$

Then

$$\mu(m) D_Q(m) = \mu(m)^2 (\Lambda(m) - J_{Q,\alpha}(m)). \quad (15)$$

If $1 \leq m < N$, $Q \geq \alpha$, and $Q(\alpha + 1) > N - 1$, this becomes

$$J_{Q,\alpha}(m) = j_\alpha(m), \quad j_\alpha(m) = - \sum_{\substack{d|m \\ d \leq \alpha}} \mu(d) \log d.$$

Proof. For complementary divisors $kd = m$, the exact identity $\mu(k)\mu(m) = \mu(m)^2\mu(d)$ holds. It follows either by multiplicativity when m is squarefree, or because both sides vanish when it is not. Substitution in (12) gives

$$\mu(m) D_Q(m) = -\mu(m)^2 \sum_{\substack{d|m \\ d > \alpha \\ m/d \leq Q}} \mu(d) \log d.$$

The full divisor sum is $-\Lambda(m)$. Its omitted part is exactly the union defining $J_{Q,\alpha}$, proving (15). If $d > \alpha$ and $m > Qd$, then $m \geq Q(\alpha + 1) + 1 > N$, contradicting $m < N$. Thus the extra condition contributes no omitted divisor under the stated hypotheses. $\square$

For the profiles used below, $\alpha = \lceil N^\vartheta \rceil$ and $Q = \lfloor (N - 1)/\alpha \rfloor$, with fixed $0 < \vartheta < 1/2$. For sufficiently large N , $Q \geq \alpha$. Writing $N - 1 = \alpha Q + r$, $0 \leq r < \alpha$, gives $Q > r$ and hence $Q(\alpha + 1) > N - 1$. This proves the required cap condition; it does not identify α with N^ϑ .

At $U = V = 1$ the actual Vaughan short component is $h_1(n) = \log n$, and $f_\Pi(n) = \Lambda(n) - \log n$ on $n > 1$. Introduce

$$\begin{aligned} \mathcal{S}_{\text{full}} &= \sum_{1 \leq n < N} \mathbf{1}_{\{(n,N)=1\}} f_\Pi(n) \mu(m) (D_Q - W_Q), \\ I &= \sum_{1 \leq n < N} \mathbf{1}_{\{(n,N)=1\}} h_1(n) \mu(m) (D_Q - W_Q), \\ J_{\text{ref}} &= \sum_{1 \leq n < N} \rho(n) \Lambda(n) \mu(m)^2 J_{Q,\alpha}(m), \\ Z_{\text{ref}} &= \sum_{1 \leq n < N} \rho(n) \Lambda(n) \mu(m) W_Q(n, m), \\ R_{\text{unit}} &= \sum_{1 \leq n < N} \rho(n) \Lambda(n) \mu(m)^2 \Lambda(m), \\ R_{\text{sf}} &= \sum_{1 \leq n < N} \mu(n)^2 \Lambda(n) \mu(m)^2 \Lambda(m). \end{aligned}$$

The last quantity is the actual weighted ordered prime-pair detector on *both* axes: its nonzero summands are $\log p \log q$, with $p + q = N$ and p, q prime.

Proposition 6.2 (Exact complete transfer). *For $U = V = 1$, with the preceding actual kernels and coefficients,*

$$\mathcal{S}_{\text{full}} = R_{\text{unit}} - J_{\text{ref}} - Z_{\text{ref}} - I. \quad (16)$$

There are nonnegative quantities P_{ret} and T_{pairs} such that $R_{\text{unit}} = R_{\text{sf}} + P_{\text{ret}} - T_{\text{pairs}}$: the former counts retained proper prime powers on the first axis with a prime second axis, and the latter counts the union of omitted actual prime pairs. For even N , put

$$E_J = \mathfrak{S}(N)N - J_{\text{ref}}, \quad E_Z = -\mathfrak{S}(N)F_N - Z_{\text{ref}}, \quad \varepsilon_{\text{tr}} = P_{\text{ret}} - T_{\text{pairs}} + E_J + E_Z - I.$$

Then

$$\mathcal{S}_{\text{full}} = R_{\text{sf}} - \mathfrak{S}(N)(N - F_N) + \varepsilon_{\text{tr}}, \quad (17)$$

$$R_{\text{sf}} = \mathfrak{S}(N)(N - F_N) - \mathcal{F}_{\text{fixed}} - P_{\text{ret}} + T_{\text{pairs}} - E_J - E_Z. \quad (18)$$

Proof. Recombine $f_{\text{II}} + h_1 = \Lambda$ and apply (15) to obtain (16). The support of $\mu(m)^2\Lambda(m)$ is precisely the prime support, so the first-axis proper powers and the omitted prime pairs give the stated nonnegative partition. This is a union partition, without double counting its low- n and nonunit faces. For even N the nonunit prime-pair face is confined to $p = q = N/2$. Substitution gives (17). Finally $\mathcal{F}_{\text{fixed}} = -(\mathcal{S}_{\text{full}} + I)$ gives (18). $\square$

Remark 6.3 (The remaining quantitative premise). If a covered decomposition with the *same complete kernels* has been proved in the form $\mathcal{S}_{\text{full}} = -E_{\text{cov}} + e + \mathcal{S}_{\text{unc}}$, then the definition $D_N = E_{\text{cov}} - \mathcal{S}_{\text{unc}} + |e|$ gives

$$D_N = -\mathcal{S}_{\text{full}} + 2\max(e, 0), \quad R_{\text{sf}} \geq \mathfrak{S}(N)(N - F_N) - D_N - \varepsilon_{\text{tr}}.$$

The existence and scope of that covered decomposition, and the desired bound on D_N , are separate obligations. No such bound is deduced from the identities. Equivalently,

$$D_N + \varepsilon_{\text{tr}} = \mathfrak{S}(N)(N - F_N) - R_{\text{sf}} + 2\max(e, 0).$$

For the complete direct identity (18), I and this artificial covered excess cancel. These are alternative accounting choices; their overlapping losses must not be added.

7 An elementary rational singular-factor enclosure

Lemma 7.1. *The infinite Euler product satisfies*

$$\frac{2541}{4096} \leq C_2 \leq \frac{11011}{16384}.$$

Consequently $\mathfrak{S}(N) \geq 2541/2048$ for every even $N \geq 2$.

Proof. For integers $Z > Y \geq 2$,

$$\prod_{j=Y+1}^Z \left(1 - \frac{1}{(j-1)^2}\right) = \frac{(Y-1)Z}{Y(Z-1)}.$$

Every factor is in $(0, 1)$. The prime tail, indexed by $j = p > Y$, is a subproduct of the all-integer tail and is therefore at least the latter. The convergence of C_2 follows from the summability of the deficits. Passing to the limit gives

$$\frac{Y-1}{Y}C_2(Y) \leq C_2 \leq C_2(Y), \quad C_2(Y) = \prod_{2 < p \leq Y} \left(1 - \frac{1}{(p-1)^2}\right).$$

Direct rational multiplication gives $C_2(13) = 11011/16384$; its $12/13$ multiple is $2541/4096$. All finite factors multiplying $2C_2$ in (10) are at least one. $\square$

A finite prime product is an upper bound for C_2 until the tail has been paid. The passage to the infinite product in this proof is written analysis; it is distinct from the finite telescoping Lean certificate. We also use the standard uniform bound

$$\mathfrak{S}(N) \leq \frac{N}{\varphi(N)} = O(\log \log N) \quad (N \text{ even}). \quad (19)$$

The first inequality follows by comparing the factors in (10) with $N/\varphi(N)$; the remaining finite subproduct of the C_2 factors bounds the full product from above. The usual totient estimate [15, Theorem 15] gives the second assertion. Numerical versions, when used, belong to the explicitly calibrated analytical appendix.

7.1 An earlier explicit reference on a restricted family

The published small-modulus bounds also give a reference estimate that does not use the reported primitive input below. Suppose $x = N - 1 \geq 8 \cdot 10^9$ and an odd prime $p \leq 97$ does not divide N . Then

$$N - F_N \geq \frac{N}{60000}. \quad (20)$$

Indeed, $1 - \mu(m) \geq 0$, and it equals 1 if $p^2 \mid m$; hence

$$N - F_N \geq N - \psi(x) + \psi(x; p^2, N).$$

Bennett–Martin–O’Bryant–Rechnitzer [11, Theorem 1.1 and Table 1] give $|\psi(x; q, a) - x/\varphi(q)| < x/(840 \log x)$ for the reduced progression $q = p^2 \leq 9409$, and the sharper coefficient $c_\psi(3) = 0.0003964$ for the two reduced classes modulo 3 at this onset. The powers of 3 contribute at most $\log x$ to $\psi(x)$. Therefore

$$N - F_N > 1 + x \left[\frac{1}{9312} - \frac{2(0.0003964) + 1/840}{\log x} - \frac{\log x}{x} \right].$$

For $x \geq 8 \cdot 10^9$, $\log x > 22$ and $\log x/x < 24/(8 \cdot 10^9)$. The remaining coefficient is at least

$$\frac{1}{9312} - \frac{2(0.0003964) + 1/840}{22} - \frac{24}{8 \cdot 10^9} = \frac{386215979}{22407000000000} > \frac{1}{60000}.$$

This proves (20). Its condition on p does not cover every even integer, and this earlier reference is not spent a second time in the later universal-profile ledger.

8 A conditional universal reference gap from missing prime squares

The universal estimate below is conditional on a reported project-derived primitive-character input. We state the exact needed input, rather than attributing its numerical constant to a different published theorem.

Reported analytic input 8.1 (Project primitive input for the reference). *Let $u = \log N \geq 8\,000\,000$. Choose the at most one possible primitive real Page pair (χ_*, β_*) among conductors at most u^{10} . For every integer $1 < r \leq u^{10}$, the complete primitive nonprincipal group satisfies*

$$\sum_{\substack{\chi \bmod r \text{ primitive} \\ \chi \neq \chi_0}} \left| \psi(N, \chi) + \mathbf{1}_{\{\chi = \chi_*\}} \frac{N^{\beta_*}}{\beta_*} \right| \leq \frac{100N}{u^8}. \quad (21)$$

When no exceptional pair exists, its correction is absent.

The retained source derives this input from the signed explicit formula, zero-free region and direct remainder estimates of Bordignon [8], including the actual Perron endpoint and the reflected exceptional-zero fee. The analytical section identifies its direct remainder envelope; the complete substitution into every printed primary-source formula has not been independently reproduced in this manuscript. It remains a reported written input, rather than a new self-contained certificate or a Lean theorem. In the application below the exceptional character is absent from the retained groups, so no model sign is assumed. A stronger uniform prefix estimate using $(y^{\beta_*} - 1)/\beta_*$ also supplies this endpoint lemma after paying the constant $1/\beta_*$ correction.

Theorem 8.2 (Conditional universal signed reference). *Assume Input 8.1. Then for every integer N with $u = \log N \geq 8\,000\,000$,*

$$N - F_N \geq \frac{N}{256u \log u}. \quad (22)$$

For even N ,

$$\mathfrak{S}(N)(N - F_N) \geq \frac{2541N}{524288u \log u}. \quad (23)$$

Proof. Put $X = N - 1$. The explicit ordinary estimates used below are Akbary–Hambrook [7, Lemma 3.1(d),(f)] and Johnston–Yang [9, Theorem 1.1]. A convenient conservative consequence of the latter is

$$|\psi(N) - N| \leq NE(u), \quad E(u) = 34u^2 \exp\left(-\frac{4}{5}\sqrt{u}\right). \quad (24)$$

Choice of the square obstructions. Consider the literal real interval $(2u, 4u]$. The explicit theta bound gives

$$\theta(4u) - \theta(2u) \geq 2u - \frac{6u}{5(\log u)^2}.$$

Remove its primes dividing N ; their logarithmic mass is at most $\log \text{rad}(N) \leq u$. Remove at most one additional prime if the possible Page conductor itself is a prime in that interval. Call the remaining set $\mathcal{P}$. This second removal costs at most $\log(4u)$, regardless of whether the pair exists. Since $\log u > 15$ and $\log(4u)/u < 1/100$ at the stated onset,

$$\sum_{\lambda \in \mathcal{P}} \log \lambda > \frac{u}{2}. \quad (25)$$

The logarithmic ratio decreases thereafter. Put

$$M = |\mathcal{P}|, \quad A = \sum_{\lambda \in \mathcal{P}} \frac{1}{\varphi(\lambda^2)}.$$

Because $\lambda \leq 4u$ and $\log(4u) \leq 2 \log u$,

$$\frac{1}{64u \log u} \leq A \leq \frac{2}{u}, \quad M \leq 4u. \quad (26)$$

Indeed (25) gives $M > u/[2 \log(4u)]$, while $\varphi(\lambda^2) \leq 16u^2$ supplies the lower bound. For the upper bound use $\lambda > 2u$, $\varphi(\lambda^2) \geq 2u^2$, and $M \leq 4u$.

The individual square progressions. For every $\lambda \in \mathcal{P}$, the class N modulo $q = \lambda^2$ is reduced and $q \leq 16u^2 \leq u^{10}$. An odd real character modulo λ^2 factors through λ : the kernel of reduction has odd order λ , whereas the real character has image of order at most two. Thus a possible primitive real conductor dividing λ^2 is λ , not λ^2 . That prime was removed if necessary. Composite or even Page conductors cannot divide these retained moduli.

Every nonprincipal character modulo λ^2 has primitive conductor λ or λ^2 . Induction between them changes no prime support, hence no von Mangoldt term. Applying Input 8.1 to the two complete groups costs at most $200N/u^8$. The principal character removes at most u in λ -power weight. Character orthogonality therefore gives

$$\psi(N; \lambda^2, N) \geq \frac{N - NE(u) - u - 200N/u^8}{\varphi(\lambda^2)}.$$

Passing to $X = N - 1$ removes the actual face $\Lambda(N) \leq u$. Since $\varphi(\lambda^2) \leq 16u^2$,

$$\begin{aligned} \psi(X; \lambda^2, N) &\geq \frac{N}{\varphi(\lambda^2)} \left(1 - E(u) - \frac{200}{u^8} - 17u^3 e^{-u} \right) \\ &\geq \frac{3N}{4\varphi(\lambda^2)}. \end{aligned} \tag{27}$$

All three error envelopes decrease from this onset, and their sum is less than $1/4$ there. These are endpoint estimates at N with the $n = N$ face paid; no altered logarithmic threshold at $N - 1$ is silently used.

Intersections and finite Bonferroni. For distinct $\lambda, \nu \in \mathcal{P}$ the intersection is the reduced class N modulo $(\lambda\nu)^2$. Its modulus is at most $256u^4 \leq N^{3/8}$. Brun–Titchmarsh at N gives the prime-log bound $16N/[5\varphi((\lambda\nu)^2)]$, because $\log(N/q) \geq 5u/8$. Grouping proper powers by their prime base gives total von Mangoldt weight at most $\sqrt{N}u$ for each intersection. CRT and $\varphi((\lambda\nu)^2) = \varphi(\lambda^2)\varphi(\nu^2)$ yield

$$\begin{aligned} \sum_{\substack{\lambda, \nu \in \mathcal{P} \\ \lambda < \nu}} \psi(X; (\lambda\nu)^2, N) &\leq \frac{8}{5}NA^2 + \frac{1}{2}M^2\sqrt{N}u \\ &\leq \frac{8}{5}NA^2 + 8\sqrt{N}u^3 \leq \frac{1}{4}NA. \end{aligned} \tag{28}$$

For the last comparison, $A \leq 2/u$ gives $(8/5)A^2 < A/8$. The remaining ratio is bounded by $512u^4 \log u e^{-u/2} < 1/8$, a decreasing envelope on this range. No exceptional-character estimate is required for these upper intersection counts.

Let G be the actual nonnegative Λ mass of the union $\{\lambda^2 \mid N - n \text{ for some } \lambda \in \mathcal{P}\}$, with $1 \leq n < N$. Finite Bonferroni and (27)–(28) give

$$G \geq \frac{1}{2}NA \geq \frac{N}{128u \log u}. \tag{29}$$

If the union indicator is one, $\mu(N - n) = 0$; otherwise $\mu(N - n) \leq 1$. Thus, pointwise, $\mu(N - n) \leq 1 - \mathbf{1}_{\text{union}}$, and $F_N \leq \psi(X) - G$. Equation (24) implies

$$N - F_N \geq G - NE(u).$$

Finally $E(u) \leq 1/[256u \log u]$ on this range. Its normalized envelope $8704u^3 \log u e^{-(4/5)\sqrt{u}}$ decreases here. At $u = 8\,000\,000$, $\sqrt{u} > 2800$, $u < 2^{23}$ and $e > 2$ already give a very generous rational comparison. The same elementary comparisons verify the previous endpoint and overlap inequalities. This proves (22). Multiplication by Lemma 7.1 proves (23) when N is even. $\square$

Remark 8.3 (What this estimate does not establish). The proof removes at most one actual prime to pay the possible Page uncertainty, and thus also treats multiples of every fixed primorial. It does not assume squarefree N or a favourable Möbius sign. Its scale $N/(\log N \log \log N)$ is smaller than a fixed positive multiple of N . It neither estimates $\mathcal{F}_{\text{fixed}}$ nor implies positivity of R_{sf} . A fixed linear exterior allowance cannot automatically be spent against this shrinking reference, and no lower-range finite coverage is supplied.

9 A negative small-part sector at the eighth-power cutoff

The next results use a *different* parameter profile:

$$U = V = 1, \quad \alpha = \lceil N^{1/8} \rceil, \quad Q = \left\lfloor \frac{N-1}{\alpha} \right\rfloor. \quad (30)$$

They have an unevaluated onset under classical Bombieri–Vinogradov. In particular the numerical onset of an exterior estimate proved with $\alpha = \lceil N^{1/4} \rceil$ must not be assigned to (30).

Definition 9.1 (The actual composite sector). Let $a = 21/500$, $b = 3/50$, and let $\mathcal{P}_N$ consist of the products $P = p_1 p_2 p_3$ of three distinct primes in $(N^a, N^b]$, none dividing N . Let $\mathcal{C}$ be the physical rows with $N/2 < n \leq 3N/4$ prime, $m = N - n = PL$, $P \in \mathcal{P}_N$, and L squarefree with every prime divisor strictly greater than α . Both parities of L are retained. Define

$$A_{\mathcal{C}} = \sum_{n \in \mathcal{C}} \Lambda(n), \quad F_{\mathcal{C}} = \sum_{n \in \mathcal{C}} \Lambda(n) \mu(m) (W_Q(n, m) - D_Q(m)).$$

The divisor and centering kernels are still (12)–(13).

Theorem 9.2 (A classical negative composite sector). *For every sufficiently large even N in (30),*

$$A_{\mathcal{C}} > \frac{N}{260 \log N}, \quad F_{\mathcal{C}} < -\frac{N}{2500}. \quad (31)$$

The threshold is unevaluated and is not asserted effective under the classical inputs used in the proof.

Proof. We give the fibre-distribution argument explicitly. In particular we do not assume Bombieri–Vinogradov separately for every P .

The exact coefficient and its uniform sign. The fixed gaps

$$3a = \frac{63}{500} > \frac{1}{8}, \quad 2b = \frac{3}{25} < \frac{1}{8}$$

imply that $P > \alpha$ eventually, whereas every proper divisor of P is at most α . The primes of L exceed α , so P is the unique entire small-prime part of m . The product m is squarefree. Also n is a unit modulo N : a prime $n > N/2$ cannot divide even N in this physical interval.

All divisors of m at most α are therefore proper divisors of P . For a squarefree product of three primes, $\sum_{d|P} \mu(d) \log d = 0$ and $\mu(P) = -1$. Removing its top divisor gives $j_{\alpha}(m) = -\log P$. The cap condition in Lemma 6.1 holds eventually. Consequently the exact actual row is

$$\Lambda(n) [-\log P + \mu(m) W_Q(n, m)]. \quad (32)$$

No pointwise sign for a character-minus-model discrepancy is being assumed here.

The true prefix R is at least $N^{7/8}/16$ eventually, since $m \geq N/4$ and $\alpha = N^{1/8} + O(1)$. The mask is the actual even integer $nN < N^2$, and the true phase is $\log(k/m)$. The fixed-polynomial-mask prefix estimate proved below gives $W_Q(n, m) = -\mathfrak{S}(nN) + o(1)$ uniformly. Since n is prime and $(n, N) = 1$,

$$\mathfrak{S}(nN) = \mathfrak{S}(N) \frac{n-1}{n-2} \leq 2\mathfrak{S}(N) = O(\log u).$$

Because $\log P > 63u/500$, the gap $63/500 - 3/25 = 3/500$ absorbs this logarithmic term. Thus, for *both* parities of the retained L ,

$$\Lambda(n) [-\log P + \mu(m) W_Q(n, m)] \leq -\frac{3u}{25} \Lambda(n) \quad (33)$$

eventually. It remains to lower-bound the actual nonnegative mass.

The weighted sequence and its local density. For $P \in \mathcal{P}_N$, put $x_P = N/P$, $X_P = N/[4\varphi(P)]$, and let $a_P(L)$ equal $\log(N - PL)$ when $N - PL$ is prime in $(N/2, 3N/4]$, and zero otherwise. Before removing squareful L , this is a nonnegative sequence supported in $L \leq x_P$. For squarefree d coprime to N , its d -divisibility count is the prime-log sum in the reduced class N modulo Pd on the same two physical endpoints. Relative to X_P , its multiplicative local density $v_P(p) = g_P(p)/p$ is

$$v_P(p) = \begin{cases} 0, & p \mid N, \\ 1/p, & p \mid P, \\ 1/(p-1), & p \nmid NP. \end{cases} \quad (34)$$

If $p \mid P$ and $p \mid d$, the actual modulus has p^2 ; $\varphi(Pp) = p\varphi(P)$ proves its displayed density. That overlap is not removed. If d has a prime divisor of N , the central prime count is exactly zero, as is the model. No nonunit class is fed into a reduced-class theorem.

For odd primes $v_P(p) \leq 1/(p-1)$, and $v_P(2) = 0$. The inverse dimension product is bounded by the ordinary odd-prime baseline. Mertens' product estimate and the convergent quadratic correction give its logarithmic-ratio upper bound with absolute constants, independently of the growing masks P, N . Thus the density condition in [13, Definition 2.4] is uniform.

Joint distribution and the good-bad partition. Take the fixed exponent $\theta_0 = 31/82$ and individual level $D'_P = x_P^{\theta_0}$. Then

$$PD'_P \leq N^{49/100}, \quad x_P \geq N^{41/50}, \quad \liminf \frac{\log D'_P}{\log(\alpha + 1)} \geq \frac{62}{25} > \frac{12}{5}. \quad (35)$$

The positive gaps absorb the integer ceilings. Define the actual error

$$E_P = \sum_{d \leq D'_P} \mu(d)^2 \left| \sum_{L \leq x_P/d} a_P(dL) - X_P \frac{g_P(d)}{d} \right|.$$

Call P good when $E_P \leq X_P/(\log x_P)^{100}$, and bad otherwise. This is an exact classification, not an assertion of individual good distribution for all fibres.

Let $E_\theta(q)$ be the maximum of $|\theta(y; q, c) - y/\varphi(q)|$ over $y \leq N$ and reduced classes $c \bmod q$. An integer $q \leq N^{49/100}$ has at most eleven distinct primes in $(N^a, N^b]$, because twelve would give $q > N^{12a} > N^{49/100}$. Each possible P is a triplet of those primes and determines $d = q/P$ uniquely. Hence there are at most $\binom{11}{3} = 165$ representations, even when q is not squarefree. Paying both actual endpoints gives

$$\sum_P E_P \leq 330 \sum_{q \leq N^{49/100}} E_\theta(q). \quad (36)$$

Classical all-prefix Bombieri–Vinogradov, in the form recalled in [7] and [14, (6.24)], bounds the corresponding ψ sum by $O_A(N/u^A)$ for every fixed A , since $N^{49/100} < \sqrt{N}/u^B$ eventually for every fixed B . The complete ψ -to- θ proper-power difference is $O(N^{99/100}u^2)$ after summing these moduli, again smaller than N/u^A for any fixed A eventually. Exceptional characters are included in this classical input; its constants and onset are not evaluated. Thus $\sum_P E_P = O_A(N/u^A)$.

Choosing $A = 300$ pays the lost bad main mass:

$$\sum_{P \text{ bad}} X_P \leq u^{100} \sum_P E_P = O(N/u^{200}), \quad \sum_{P \text{ bad}} \frac{1}{\varphi(P)} = O(u^{-200}). \quad (37)$$

Only the nonnegative actual count is discarded on bad fibres in the lower estimate. Their model errors, or signed discrepancies, are not called positive. Each good sequence meets [13,

Definition 2.4] with distribution constant one and the same uniform density constants. Its Lemma 2.5 therefore supplies a relative $o(1)$ uniformly in P, N ; its constants do not depend on the particular physical sequence.

The main mass and its constants. Write $z = \alpha + 1$ so that primes $p \leq \alpha$ are excluded. The baseline product satisfies

$$V_0(z) = \prod_{2 < p < z} \left(1 - \frac{1}{p-1}\right) \sim \frac{2e^{-\gamma}C_2}{\log z}.$$

The true product $V_P(z) = \prod_{p < z} (1 - v_P(p))$ is at least $V_0(z)$. The lower sieve function in [13, (2.4)–(2.5)] is $f(s) = 2e^\gamma \log(s-1)/s$ for $2 \leq s \leq 4$. The gap in (35) and monotonicity on the present interval give

$$\liminf u f(s_P) V_P(z) \geq \frac{40}{3} C_2 \log \frac{7}{5} > \frac{40}{9} C_2 > \frac{27}{10}. \quad (38)$$

Here $s_P = \log D'_P / \log z$. The strict logarithm inequality $\log(7/5) > 1/3$ follows, for example, from $e < 11743/4320 < 49/18 < (7/5)^3$. This rational upper bound for e is obtained by summing through $1/5!$ and bounding the remaining tail by $7/4320$. Lemma 7.1 proves the last rational comparison. The uniform relative sieve error is absorbed *before* using $V_P \geq V_0$.

The reciprocal-prime mass in the window tends to $\log(b/a) = \log(10/7) > 1/3$. At most twenty-three of its primes divide N , and their removed reciprocal mass is at most $24N^{-a}$. The equal-prime diagonals tend to zero. The unordered distinct triples therefore have limiting reciprocal mass greater than $1/162$. Since $1/\varphi(P) \geq 1/P$, subtracting the paid bad mass in (37) leaves

$$\sum_{P \text{ good}} \frac{1}{\varphi(P)} > \frac{1}{170} \quad (39)$$

eventually, uniformly over even N . The lower sieve on the good fibres consequently supplies rough- L prime-log mass at least $(27/6800 - o(1))N/u$.

Squarefree L and the actual signed sum. For each P , an offending squared prime $p > \alpha$ contributes at most $N/(Pp^2) + 1$ possible integers L . Keeping this $+1$, the total lost prime-log mass is at most

$$u \left[\frac{N}{\alpha} \sum_P \frac{1}{P} + \sum_P \sqrt{\frac{N}{P}} \right] = O(N^{7/8}u) + O(N^{617/1000}u) = o(N/u). \quad (40)$$

We used the bounded P harmonic mass, $P > N^{63/500}$ and at most $N^{9/50}$ distinct integer products P . The exponent in the second term is $1/2 + 9/50 - 63/1000 = 617/1000$. The remaining original summands have unique P and all their original masks. The combined qualitative losses are eventually less than $N/(10000u)$. Since $27/6800 - 1/10000 > 1/260$, this proves $A_C > N/(260u)$. Applying (33) gives $F_C < -3N/6500 < -N/2500$. $\square$

Remark 9.3 (Scope of the lower sieve). Only P , the small part of m , has three prime factors. The cofactor L may have further rough factors, with both parities retained; the theorem is not a prime-plus-three-primes theorem. The good–bad argument uses classical BV with an unevaluated, ineffective threshold. The rational parameter checks do not evaluate it. The proof uses the normalized primary lower sieve of [13]; it does not adopt a preliminary argument based on the erroneous printed remainder sign in Richert’s (11.80).

10 Uniform prefixes and the entire rough complement

We record the polynomial-mask prefix estimate needed both in the preceding theorem and in the rough complement. It is a written ordinary Mertens consequence; no shifted-prime cancellation is assumed.

Lemma 10.1 (A fixed-polynomial-mask prefix). *Let K be even and positive, and define*

$$A_K(x) = \sum_{\substack{r \leq x \\ (r, K)=1}} \frac{\mu(r)}{\varphi(r)}, \quad W_K(R) = \sum_{\substack{r \leq R \\ (r, K)=1}} \frac{\mu(r)}{\varphi(r)} \log \frac{r}{R}.$$

There is an absolute c such that, for every fixed $D > 1$,

$$|A_K(x)| \ll_D \frac{\log(2K)}{\log(2x)^D} + x^{-1/4} \exp(c\sqrt{\log(2K)}),$$

$$|W_K(R) + \mathfrak{S}(K)| \ll_D \frac{\log(2K)}{\log(2R)^{D-1}} + R^{-1/4} \exp(c\sqrt{\log(2K)}).$$

For every fixed C , these save any prescribed logarithmic power uniformly when $K \leq R^C$. For the actual phase one has exactly

$$\sum_{\substack{r \leq R \\ (r, K)=1}} \frac{\mu(r)}{\varphi(r)} \log \frac{r}{m} = W_K(R) + \log \frac{R}{m} A_K(R). \quad (41)$$

Proof. Put $a_K(r) = \mu(r)\mathbf{1}_{\{(r, K)=1\}}/\varphi(r)$ and $b(r) = \mu(r)/r$. The exact convolution $a_K = h_K * b$ has

$$h_K(p^j) = \begin{cases} p^{-j}, & p \mid K, \\ -p^{-j}/(p-1), & p \nmid K, \end{cases} \quad j \geq 1.$$

Its absolute Euler products give

$$\sum_d |h_K(d)| \ll \log(2K), \quad \sum_d |h_K(d)|\sqrt{d} \ll \exp(c\sqrt{\log(2K)}).$$

The first product is the reciprocal-totient product on the mask times an absolutely convergent quadratic correction. The second is bounded by splitting its mask primes at a logarithmic scale; the remaining prime mass is limited by $\log K$. These are uniform absolute moments, not fixed- K unspecified constants. The ordinary phase-zero Mertens estimate and partial summation give $B(t) = \sum_{j \leq t} \mu(j)/j \ll_D \log(2t)^{-D}$, with total harmonic sum zero. Such an ordinary input follows, for example, from [10]; no effectivity of the final classical lower-sieve threshold is claimed here.

In $A_K(x) = \sum_{d \leq x} h_K(d)B(x/d)$ split at $\sqrt{x}$. The first moment handles the head, and boundedness of B with the half moment gives the tail $x^{-1/4} \sum |h_K(d)|\sqrt{d}$. Moreover $W_K(R) = -\int_1^R A_K(t) dt/t$. For each fixed K , its convergent full integral is the value $\mathfrak{S}(K)$, obtained from $\zeta(1+s)^{-1}H_K(s)/s$ as $s \downarrow 0$. The absolutely convergent H_K at zero gives $H_K(0) = \mathfrak{S}(K)$ for even K by its Euler factors. Subtracting the finite integral and applying the already uniform tail estimate yields the second bound. Thus a fixed- K limit has not been used as an unproved uniform limit. Finally (41) is the exact finite logarithmic identity. Its endpoint term must be retained. $\square$

Theorem 10.2 (A one-sided upper for all rough rows). *On (30), let F_{rough} be the complete physical contribution to (14) with squarefree $m = N - n$ and all prime factors of m greater than α . Include $m = 1$ with its zero value. Uniformly over even N ,*

$$F_{\text{rough}} \leq o(N). \quad (42)$$

This is a one-sided upper, not an absolute-value estimate.

Proof. Literal completion and the actual prefix. If m is α -rough, every proper divisor $k \mid m$ has $m/k > \alpha$ and $k < m/\alpha \leq (N-1)/\alpha$. Its integer k is therefore at most Q , and the strict boundary admits it. The omitted $k = m$ term has zero logarithm. Full divisor convolution gives $D_Q(m) = -\Lambda(m)$ exactly.

For every rough composite m , $m > \alpha^2 \geq N^{1/4}$. The same lower endpoint can be used for rough prime $m \geq N^{1/4}$. Since $\alpha \leq 2N^{1/8}$, the exact integer prefix has $R \geq N^{1/8}/8$ eventually. Hence the actual even mask $K = nN < N^2$ satisfies $K \leq R^{18}$ eventually, while $|\log(R/m)| \leq u = O(\log R)$. Lemma 10.1 and its *shifted* identity give

$$W_Q(n, m) = -\mathfrak{S}(nN) + o(1) \quad (43)$$

uniformly on these rows. The safe fixed exponent 18 absorbs the factor 8 in the endpoint; $K \leq R^{16}$ does not follow automatically from this lower bound. No previously evaluated prefix theorem requiring $R \geq N^{1/5}$ is invoked.

Whenever a unit first-axis prime power $n = p^j$ contributes, p is odd, $p \nmid N$ and

$$\mathfrak{S}(nN) = \mathfrak{S}(N) \frac{p-1}{p-2} \leq 2\mathfrak{S}(N).$$

Equations (19) and (43) show that $W_Q < 0$ and $|W_Q| \leq 3\mathfrak{S}(N)$ eventually.

Favourable signs and the corner budgets. For a rough prime $m \geq N^{1/4}$, the actual row is $\Lambda(n)[- \log m - W_Q] \leq 0$ eventually. For a rough squarefree composite with even $\Omega(m)$ it is $\Lambda(n)W_Q \leq 0$. Odd composite parity contributes $-\Lambda(n)W_Q \leq 3\mathfrak{S}(N)\Lambda(n)$. Eight primes greater than α would force $m > N$; thus the only positive composite parities are 3, 5, 7.

We pay the exceptional faces absolutely without changing their support. For every $k \geq 1$, $\varphi(k) \geq \sqrt{k/2}$: in $\varphi(k)^2/k$ every prime-power factor is at least one except the single possible factor $p = 2$, exponent one, which equals $1/2$. Consequently

$$\sum_{k \leq Q} \frac{1}{\varphi(k)} \leq 2\sqrt{2Q}, \quad |W_Q - D_Q| \leq u(1 + 2\sqrt{2Q}) = O(uN^{7/16}).$$

Grouping all proper first-axis prime powers by their base gives total von Mangoldt mass at most $\sqrt{N}u$. Their complete contribution is $O(N^{15/16}u^2) = o(N)$. Prime $m < N^{1/4}$ costs $O(N^{11/16}u^2)$, and prime $n \leq N^{1/8}$ costs $O(N^{9/16}u^2)$. Overlap among these nonnegative upper allowances is harmless. These are single-Mangoldt corner budgets, not another two-axis detector fee or a second Type-I allowance. Nonsquarefree m already contributes zero.

A single finite upper sieve for the positive union. Put $z = N^{1/8}$. For the remaining prime $n > z$ and rough m , both forms $n, N - n$ avoid every prime below z . Let $B(N, z)$ count all integers $1 \leq n < N$ with this property. The root number for $n(N - n)$ is

$$\omega(p) = \begin{cases} 1, & p \mid N, \\ 2, & p \nmid N. \end{cases}$$

In particular $\omega(2) = 1$. Put $g(p) = \omega(p)/(p - \omega(p))$ and $G(z) = \sum_{d < z} \mu(d)^2 g(d)$. The finite Selberg identities [14, (9.28)–(9.33)] give optimal weights with $\lambda_1 = 1$, $|\lambda_d| \leq 1$ and quadratic main $1/G(z)$. CRT has remainder at most $\omega([d, e]) \leq [d, e] \leq de$. The full double remainder is therefore at most z^4 , giving

$$B(N, z) \leq \frac{N}{G(z)} + z^4. \quad (44)$$

This is a count of all integers in two forms, with no prime-AP distribution assumption.

For completeness, the needed lower G bound is uniform in the growing mask. Take $y = z^{1/16}$ and the finite positive Euler mass $\mathcal{E}(y) = \prod_{p \leq y} (1 - \omega(p)/p)^{-1}$. Under its divisor probability,

the mean of $\log d$ is $\sum_{p \leq y} \omega(p) \log p/p \leq 4 + 4 \log y$, using the published global $\theta(t) \leq 2t$ bound in [7]. For $\log z \geq 16$, this is at most $\frac{1}{2} \log z$; Markov gives $G(z) \geq \mathcal{E}(y)/2$. The exact Euler factors, including the factor at two, imply

$$\mathcal{E}(y) \geq \frac{C_2}{\mathfrak{S}(N)} \left[\prod_{p \leq y} (1 - 1/p)^{-1} \right]^2.$$

The full identity has the extra factors $1/C_2(y) \geq 1$ and $\prod_{p|N, p > y} (p-1)/(p-2) \geq 1$; they are not discarded in an adverse direction. The harmonic Euler product is at least $\sum_{j \leq [y]} 1/j \geq \log y$. Lemma 7.1 thus gives

$$G(z) \geq \frac{2541}{2097152 \mathfrak{S}(N)} (\log z)^2, \quad B(N, z) \leq \frac{134217728 \mathfrak{S}(N) N}{2541 u^2} + \sqrt{N}.$$

The entire positive parity union 3, 5, 7 is counted *once*. Since prime first-axis weights are at most u , its contribution and the paid corners satisfy

$$F_{\text{rough}} \leq \frac{402653184 \mathfrak{S}(N)^2 N}{2541 u} + 3 \mathfrak{S}(N) u \sqrt{N} + o(N) = o(N),$$

because $\mathfrak{S}(N) = O(\log u)$. The favourable rough prime and even-parity contributions have been retained, not asserted absolutely small. This proves (42). $\square$

Corollary 10.3 (A disjoint combined negative sector). *On (30), for every sufficiently large even N ,*

$$F_{\mathcal{C}} + F_{\text{rough}} < -\frac{N}{3000}. \quad (45)$$

The onset remains unevaluated and ineffective under the classical lower-sieve input.

Proof. The sector $\mathcal{C}$ has a nontrivial α -small part, so it is disjoint from the all-rough sector. Eventually the one-sided upper in Theorem 10.2 is less than $N/15000$. Theorem 9.2 and $1/2500 - 1/15000 = 1/3000$ give the claim. $\square$

Remark 10.4 (The unestimated complementary scalar). The exact partition is

$$\mathcal{F}_{\text{fixed}} = F_{\mathcal{C}} + F_{\text{rough}} + F_{\text{nonrough,rem}}.$$

There is no sufficient upper bound for $F_{\text{nonrough,rem}}$ in this work. In particular positive one-small-prime rows are not removed. The negative j_α credit is already inside J_{ref} and cannot be spent a second time after replacing the *entire* J_{ref} by its main. The new eighth-power profile has not inherited the evaluated exterior estimates of a quarter-power profile. Neither this corollary nor the universal reference theorem proves a bound on D_N or an effective positive prime-pair margin.

11 Relation to the finite formal ingredients

The manuscript distinguishes finite identities from their analytical uses. The actual Vaughan fibre identity $\mu * \beta_V = \Lambda \mathbf{1}_{\{n > V\}}$, literal cap and coprimality identities, character decompositions and finite dispersion partitions have separate Lean certificates. Complete fibre cancellation is not assumed on an incomplete selected fibre. The exact prime support of $\mu(n)^2 \Lambda(n)$ likewise supplies the finite detector interpretation in Proposition 6.2. The infinite Euler limits, ordinary Mertens bounds, prime-AP estimates, linear sieve and asymptotic conclusions above are written analysis; they have not been formalized in Lean.

Two earlier conventions are particularly relevant to the use of upper-sieve and bilinear reductions. If $I_W(n)$ denotes absence of prime factors at most W and $\Phi(n, z) = \mathbf{1}_{\{(n, \prod_{p < z} p) = 1\}}$, then for integer W one has $I_W = \Phi(\cdot, W + 1)$, while for real W the exact endpoint is $\lfloor W \rfloor + 1$. The stopped term in Rosser's identity [16, Lemma 8] uses $\Phi(n, p(e))$, not $I_{p(e)}(n)$, which vanishes whenever $e \mid n$. Its fundamental-lemma bound is an additional analytical input, not the finite primordial-boundary certificate.

Similarly, completion of selected bilinear or HH roots cannot erase their actual support or alter the modulus ar to a proper sign carrier $T \mid ar$. A small continuous density main or a local positive amplitude is not a bound for the discrete signed observable. These cautions explain why the sector inequalities above preserve the actual scalar and identify its remaining complement rather than announce a global cancellation theorem.

The source audits and isolated compilation replays were performed within an AI-assisted workflow. They are internal reproducibility checks, not external peer review. No claim of external priority, a new general sieve theorem, a completed parity-barrier argument, or a proof of binary Goldbach is made here.

12 Explicit exterior estimates and their analytic scope

This section records the analytic component of the project. Its finite coefficient identities, written analytic deductions, published inputs and numerical certificates have different verification statuses. In particular, none of the complex analysis in this section is claimed to have been formalized in Lean. Independent agent reviews are useful reproducibility evidence, but are not external mathematical peer review.

There is also a distinction between a detailed proof and a reported analytic certificate. The ordinary Mertens transport, the coefficient moments, the finite growing-mask estimate and the smoothed contour argument are explained below. The direct substitution in the long explicit remainder formula of Bordignon is recorded as an explicit reported analytic input in Lemma 12.1. The final numerical assembly is conditional on that certificate and on the stated literal completion estimates. This qualification prevents a compilation receipt or a finite rational test from being substituted for an unprovided infinite-range analytic proof.

12.1 Parameters, genuine weights and historical allowances

All logarithms are natural. Write

$$u = \log N, \quad \ell = \log u, \quad L = 1 + u, \quad X = N - 1,$$

and, for the quarter-power profile, use

$$\alpha = \lceil N^{1/4} \rceil, \quad Q = \left\lfloor \frac{N - 1}{\alpha} \right\rfloor, \quad B = \lceil L^{20} \rceil, \quad Y = B^2 \alpha, \quad R = \lfloor L^{16} \rfloor.$$

Unless otherwise stated, N is even and $1 \leq V \leq N^{1/4}$. The Type-I bounds below further require $U = V = 1$. On this specialization the genuine Vaughan coefficient is

$$\beta_1(n) = \sum_{d \mid n, d > 1} \Lambda(d) = \log n.$$

It is not an arbitrary coefficient with the same size. At $u \geq 10^6$, the rounding guards give

$$B \leq 2L^{20}, \quad \alpha \leq 2N^{1/4}, \quad Y \leq 8L^{40}N^{1/4} \leq N^{3/8}, \quad R \geq L^{16}/2.$$

For example, $\log 8 < 3$ and $\log L \leq \sqrt{L}$ reduce the third inequality to $3 + 40\sqrt{1+u} \leq u/8$; its left-to-right ratio decreases after the displayed onset.

The finite, repeated-modulus coefficient is

$$w_B(q) = \sum_{\substack{b \leq B, d \leq \alpha \\ [b^2, d] = q, (bd, N) = 1}} \mu(b)\mu(d) \log d. \quad (46)$$

Each active q is odd, coprime to N , and cube-free. Its unique factorization $q = b^2c$, with b, c square-free and $(b, c) = 1$, gives

$$w_B(b^2c) = \mu(b)\mu(c) \sum_{\substack{e|b \\ ce \leq \alpha}} \mu(e) \log(ce). \quad (47)$$

The constraint $ce \leq \alpha$ is literal. A complete-fibre identity cannot be used when $bc > \alpha$.

The following allowances describe different historical evaluations of the same components. They are not cumulative errors.

Quantity	Written allowance	Stated onset in $u = \log N$
R_{main} , finite reference model	$10^{-30}N$	10^6
R_{main} , refined evaluation	$10^{-40}N$	$8 \cdot 10^6$
Whole E_J , all-prefix evaluation	$10^{-6}N$	$8 \cdot 10^6$
Whole I , shrinking evaluation	$N/(10^{12}u\ell)$	2^{128}
Whole E_J , shrinking evaluation	$N/(10^{12}u\ell)$	2^{128}
Whole E_Z , shrinking evaluation	$N/(10^{100}u\ell)$	2^{128}
Adaptive exterior assembly	see (72)	10^{24}

These are bounds for exterior components; their onsets are not Goldbach thresholds. The qualitative sector with $\alpha = \lceil N^{1/8} \rceil$ elsewhere in this manuscript has a different profile and no evaluated onset. Its conclusions cannot be imported into this table.

12.2 Ordinary Mertens input and exact harmonic transport

We pin the ordinary Mertens input to Lee–Leong [10], arXiv:2208.06141v5, dated 9 September 2026. The constants used belong to the classical block of their table, with

$$c = \frac{839}{25} = 33.56, \quad w = \frac{139}{25} = 5.56, \quad s_0 = \frac{22946}{5} = 4589.20.$$

Their published theorem supplies

$$|M(t)| \leq ct \log t \exp\{-\sqrt{\log t/w}\}, \quad \log t \geq s_0, \quad M(t) = \sum_{n \leq t} \mu(n). \quad (48)$$

The computation underlying the published constant is an external input, not a computation revalidated by this manuscript. The unversioned 2022 preprint must not be cited as though all its constants coincided with v5.

Put $B_\mu(t) = \sum_{n \leq t} \mu(n)/n$. A useful global short-argument bound is $|B_\mu(t)| \leq 1$ for $t \geq 1$. Indeed, for $m = \lfloor t \rfloor$, Dirichlet inversion gives

$$\sum_{n \leq m} \mu(n) \lfloor m/n \rfloor = 1, \quad mB_\mu(m) = 1 + \sum_{n \leq m} \mu(n) \{m/n\}.$$

The $n = 1$ fractional part vanishes and there are at most $m - 1$ remaining summands of absolute value at most one. Partial summation and (48) show that $B_\mu(t)$ converges. Its limit is zero: for $\eta > 0$, absolute Dirichlet convolution gives $\sum \mu(n)n^{-1-\eta} = 1/\zeta(1+\eta)$, while partial summation gives its Abel limit as the limit of B_μ . The inequality $\zeta(1+\eta) \geq 1/\eta$ forces that limit to be zero. Consequently,

$$B_\mu(t) = \frac{M(t)}{t} - \int_t^\infty \frac{M(v)}{v^2} dv.$$

For $s = \log t \geq s_0$ and $z = \sqrt{s/w}$ this proves

$$|B_\mu(t)| \leq ce^{-z}\{s + 2w^2P_3(z)\}, \quad P_3(z) = z^3 + 3z^2 + 6z + 6. \quad (49)$$

Here the integral is obtained by substituting $\log v = wz^2$ and integrating z^3e^{-z} three times. Omitting the integral would not give the stated harmonic-prefix bound.

For a positive integer K let

$$a_K(r) = \frac{\mu(r)}{\varphi(r)} \mathbf{1}_{(r,K)=1}.$$

Division of local Euler factors yields the exact convolution $a_K = h_K * (\mu(r)/r)$, where h_K is multiplicative and

$$h_K(p^j) = \begin{cases} p^{-j}, & p \mid K, \\ -p^{-j}/(p-1), & p \nmid K, \end{cases} \quad j \geq 1, \quad h_K(1) = 1.$$

The absolutely convergent products give the uniform moments

$$\sum_d |h_K(d)| \leq 8(1 + \log K), \quad \sum_d \sqrt{d} |h_K(d)| \leq 32 \exp\{6\sqrt{\max(2, \log(2K))}\}. \quad (50)$$

For the first bound, the unmasked product is at most $2\exp(1/2) < 4$; the masked product is at most $\zeta(2) \sum_{d \mid \text{rad } K} 1/d \leq 2(1 + \log K)$. For the second, isolate the factors at 2, 3, bound the rest by $\exp\{(5/2) \sum_{p \geq 5} p^{-3/2}\} < 5$, and use $-\log(1 - p^{-1/2}) \leq 2p^{-1/2}$ on the masked factors. Splitting the prime divisors of K at $Y_0 = \max(2, \log(2K))$ bounds their reciprocal-square-root sum by $3\sqrt{Y_0}$. These estimates retain every prime of the growing mask.

Define

$$A_K(x) = \sum_{r \leq x} a_K(r), \quad W_K(R) = \sum_{r \leq R} a_K(r) \log(r/R).$$

For $s \geq 2s_0$ set $z = \sqrt{s/(2w)}$ and

$$\begin{aligned} \Phi_A(s) &= ce^{-z}\{wz^2 + 2w^2P_3(z)\}, \\ \Phi_W(s) &= ce^{-z}\{4w^2P_3(z) + 8w^3Q_4(z)\}, \\ Q_4(z) &= z^4 + 7z^3 + 27z^2 + 60z + 60. \end{aligned}$$

Split the convolution at $d = \sqrt{x}$ and use the global short bound on its tail. With $L_0 = \sum |h_K|$ and $L_{1/2} = \sum \sqrt{d} |h_K(d)|$,

$$|A_K(x)| \leq L_0 \Phi_A(\log x) + L_{1/2} x^{-1/4}, \quad (51)$$

$$|W_K(R) + H_K(0)| \leq L_0 \Phi_W(\log R) + 4L_{1/2} R^{-1/4}, \quad (52)$$

where $H_K(0) = \sum h_K(d)$. To justify the nonzero main term, $W_K(R) = -\int_1^R A_K(t) dt/t$ exactly. Absolute integrability on the large tail, the Dirichlet series $H_K(s)/\zeta(1+s)$ and the residue-one

pole of ζ imply $\int_1^\infty A_K(t)dt/t = H_K(0)$. For even K , this value is $\mathfrak{S}(K)$; for odd K it is zero. The integral giving (52) is verified by $Q_4 - Q'_4 = zP_3$. The real phase is

$$\sum_{r \leq R} a_K(r) \log(r/m) = W_K(R) + \log(R/m)A_K(R). \quad (53)$$

Thus the endpoint term must be retained. An empty $R = 0$ prefix is zero and has no limiting main term.

Using $P_3(z) \leq 16z^3$, $Q_4(z) \leq 155z^4$, $c < 34$, $w < 6$, (50) implies the useful explicit envelope

$$u\{|W_K(R) + H_K(0)| + u|A_K(R)|\} \leq 4 \cdot 10^8 u^5 e^{-\sqrt{u/60}} + 160u^2 e^{-u/40} \quad (54)$$

when $K \leq N^3$, $R \geq N^{1/5}$ and $u \geq 10^6$. Indeed, the head has $z \geq \sqrt{u/60}$ and the half-moment tail has exponent $12\sqrt{u} - u/20 \leq -u/40$. These statements use the actual quotient arguments and do not invoke an asymptotic notation with an unevaluated constant.

12.3 The finite model and the signed conductor projection

Write $T = X - V$ and

$$C_B = - \sum_q \frac{w_B(q)}{\varphi(q)}, \quad R_{\text{main}} = \mathfrak{S}(N)N - TC_B.$$

For $p \nmid N$ put $a_p = 1/[p(p-1)]$. Completing the b sum first, at each fixed square-free d , gives a Dirichlet series

$$F_1(s) = \prod_{p \nmid N} (1 - a_p - p^{-1-s}) = \frac{H_1(s)}{\zeta(1+s)}, \quad \Re s > 0.$$

For an odd square-free conductor r coprime to N , forcing $r \mid [b^2, d]$ changes the local factor at $p \mid r$ to $-a_p - p^{-1-s}$. Hence

$$F_r(s) = \prod_{p \nmid Nr} (1 - a_p - p^{-1-s}) \prod_{p \mid r} (-a_p - p^{-1-s}) = \frac{H_r(s)}{\zeta(1+s)}.$$

The quotient of the changed and unchanged factors at $s = 0$ is $-1/(p-2)$. Consequently

$$H_r(0) = \mathfrak{S}(N)\kappa(r), \quad \kappa(r) = \frac{\mu(r)}{\prod_{p \mid r} (p-2)}. \quad (55)$$

This κ is signed. It differs from the positive coefficient g in the Type-I divisor expansion below.

The coefficients of H_r have local factors

$$\begin{aligned} p \mid N : & \quad (1 - z/p)^{-1}, \\ p \nmid Nr : & \quad 1 - a_p - a_p \sum_{j \geq 1} z^j/p^j, \\ p \mid r : & \quad -a_p - (1 + a_p) \sum_{j \geq 1} z^j/p^j, \end{aligned} \quad z = p^{-s}.$$

Their coefficient at 1 is not generally 1. The unmasked absolute zeroth-moment factors are at most 1, and the forced factor is $p/(p-1)^2 \leq 1$ for $p \geq 3$. Their half-moment factors are at most 1 for $p \geq 5$, since $p(1 - a_p)^2 \geq 5(19/20)^2 > 4$; the factor at 3 is less than 2. Thus, uniformly in r ,

$$\sum |h_r(d)| \leq 1 + 2u, \quad \sum \sqrt{d}|h_r(d)| \leq 2e^{12\sqrt{u}}.$$

This gives a conductor-uniform harmonic transport; it does not assert that the singular projection itself is zero.

A conservative weakening of (48) is $|M(t)| \leq 34t \log t e^{-\sqrt{\log t}/6}$ for $\log t \geq 5000$. Repeated partial summation gives, with $v = \sqrt{\log t}$,

$$\begin{aligned}\beta(v) &= 34e^{-v/6}(12v^3 + 217v^2 + 2592v + 15552), \\ \omega(v) &= 68e^{-v/6}(72v^4 + 3030v^3 + 70092v^2 + 934416v + 5606496).\end{aligned}$$

They bound $|B_\mu(t)|$ and $|\sum_{n \leq t} \mu(n)n^{-1} \log(n/t) + 1|$, respectively. The polynomial identities $(B_0 - v^2)' - (B_0 - v^2)/6 = -2v^3$ and $T'_0 - T_0/6 = -vB_0$ verify the integrations. Splitting the new coefficient convolution at $\sqrt{\alpha}$ retains its coefficient tail and its $\log \alpha$ endpoint. The resulting error is at most

$$\begin{aligned}\mathcal{H}(u) &= 9(1 + 2u)[\omega(\sqrt{u/8}) + u\beta(\sqrt{u/8})], \\ \mathcal{G}(u) &= (1 + 2u + 2u^2)e^{16+12\sqrt{u}-u/16}.\end{aligned}$$

The infinite model $b > B$ tail is distinct from a physical prime progression tail. Put $b = gh, d = gk$, where $g = (b, d)$; then $[b^2, d] = g^2 h^2 k$. Taking absolute values before removing masks gives

$$|\text{model tail}| \leq \frac{3uL}{B}[L(6 + 2 \log B) + 4(\log B)^2 + 16 \log B + 16] \leq 7000L^{-16}.$$

For example, $\sum_{gh > t} (gh)^{-2} \leq (6 + 2 \log t)/t$ follows from the harmonic hyperbola decomposition; integrating this estimate gives the log-weighted tail. The factor 3 in $q/\varphi(q)$ is already included in 7000 and is not charged again.

For $M_r(B) = \sum_{r|q} w_B(q)/\varphi(q)$ this proves the reported finite projection envelope

$$|M_r(B) + \mathfrak{S}(N)\kappa(r)| \leq \mathcal{H}(u) + \mathcal{G}(u) + 7000L^{-16}. \quad (56)$$

The same calculation at $r = 1$, together with the exact endpoint $N - T = V + 1$, gives

$$\frac{|R_{\text{main}}|}{N} \leq \mathcal{H}(u) + \mathcal{G}(u) + 7000L^{-16} + 4Le^{-3u/4}. \quad (57)$$

At $u \geq 8 \cdot 10^6$ these displayed envelopes were evaluated in the project below 10^{-40} . At $u \geq 8 \cdot 10^{10}$ each of $\mathcal{H}, \mathcal{G}, 7000L^{-16}, 4Le^{-3u/4}$ is at most u^{-4} . For instance, with $v = \sqrt{u/8}$, $u^4 \mathcal{H}(u) < 2^{49} v^{15} e^{-v/6}$; at the latter onset $v = 10^5$, the positive factor is less than 2^{304} and the negative exponential is less than 2^{-16000} . The envelope decreases because $15/v - 1/6 < 0$. Similar comparisons give $u^4 \mathcal{G} \leq 5u^6 e^{-u/32}$ and the two elementary polynomial and endpoint bounds. This variable evaluation is needed when the reference itself decreases like $N/(u \log u)$.

An active real primitive exceptional character has odd square-free conductor. Indeed, odd prime-power unit kernels have odd order, so a real character factors through the prime modulus locally. The active moduli are odd and unit against N . Even or nonunit conductors have zero actual projection. This is a statement about real primitive characters, not a claim that every primitive conductor is square-free.

12.4 Arithmetic moments and all-module conductor transport

By (47), $|w_B(b^2 c)| \leq \tau(b) \log \alpha$ and $\varphi(b^2 c) = b\varphi(b)\varphi(c)$. Dropping masks only after a positive majorization gives

$$\sum_q \frac{|w_B(q)|^2}{\varphi(q)} \leq (\log \alpha)^2 \prod_{p \geq 3} \left(1 + \frac{4}{p(p-1)}\right) \sum_{\substack{c \leq \alpha \\ c \text{ odd}}} \frac{1}{\varphi(c)}.$$

The product is at most $e^2 < 9$. From $c/\varphi(c) = \sum_{d|c} \mu(d)^2/\varphi(d)$, the odd reciprocal totient sum is at most $e^{1/2}(1 + \log \alpha) < 2(1 + \log \alpha)$. Consequently, including every truncated fibre,

$$\sum_q \frac{|w_B(q)|^2}{\varphi(q)} \leq 18L^3. \quad (58)$$

Let $E(q)$ be the maximum over all reduced classes and all $1 \leq y \leq N$ of $|\psi(y; q, a) - \psi(y)/\varphi(q)|$. Brun–Titchmarsh, in the form quoted in [7], gives a prime contribution at endpoint N at most $16N/(5\varphi(q))$ for $q \leq N^{3/8}$. Proper prime powers cost at most $\sqrt{N}u \leq N^{5/8} \leq N/\varphi(q)$. Using monotonicity of the positive prefixes, not Brun–Titchmarsh at an inadmissibly short argument, proves $E(q) < 7N/\varphi(q)$. Thus

$$\sum_q |w_B(q)|^2 E(q) \leq 126NL^3. \quad (59)$$

For a fixed Page pair (χ_*, r_*, β_*) define

$$H_{\beta_*}(y) = \frac{y^{\beta_*} - 1}{\beta_*} = \int_1^y t^{\beta_*-1} dt, \quad \Delta^\#(y; q, a) = \psi(y; q, a) - \frac{\psi(y)}{\varphi(q)} + \mathbf{1}_{r_*|q} \frac{\chi_*(a)H_{\beta_*}(y)}{\varphi(q)}.$$

If there is no pair the added term is zero. Since $0 \leq H \leq y - 1$, the corresponding maximum $E^\#(q)$ satisfies

$$E^\#(q) \leq \frac{8N}{\varphi(q)}, \quad \sum_q |w_B(q)|^2 E^\#(q) \leq 144NL^3. \quad (60)$$

The high-conductor input is Theorem 1.2 of [7]. It states

$$\mathcal{M}(t) := \sum_{d \leq t} \frac{d}{\varphi(d)} \sum_{\chi \pmod{d}}^* \max_{y \leq N} |\psi(y, \chi)| \leq c_0 u^{7/2} [4N + 2\sqrt{N}t^2 + 6N^{2/3}t^{3/2} + 5N^{5/6}t],$$

where the published formula permits $c_0 < 50$. A rational check uses $693/1000 < \log 2 < 694/1000$, $\log(4/3) > 287/1000$, $\log(\log 2 / \log(4/3)) < 89/100$, $\pi > 157/50$, $\sqrt{2} < 283/200$ and $\sqrt{\psi(113)/113} < 51/50$; substitution in that formula gives $2567195785600/52251571449 < 50$. The identification of its maximum with $\psi(113)/113$ remains a published input.

The exchange $q = dm$, $\varphi(dm) \geq \varphi(d)\varphi(m)$ and $\sum_{m \leq z} 1/\varphi(m) < 3(1 + \log z)$ give, on all modules,

$$\begin{aligned} H_{>R} &\leq 3L \sum_{R < d \leq Y} \frac{A(d)}{\varphi(d)} \\ &\leq 3c_0 L u^{7/2} [4N/R + 4\sqrt{N}Y + 18N^{2/3}\sqrt{Y} + 5N^{5/6}(1 + \log(Y/R))] \\ &\leq 5250NL^{-23/2}, \end{aligned}$$

where $A(d) = \sum_{\chi}^* \max_{y \leq N} |\psi(y, \chi)|$. The second line is Stieltjes summation of $\mathcal{M}(t)/t$; only favorable endpoint terms are dropped. The normalized bracket is at most $8L^{-16} + 4e^{-u/8} + 18e^{-7u/48} + 5Le^{-u/6} \leq 35L^{-16}$. This use of AH1.2 does not impose a least-prime condition. AH1.3 would have such a condition and cannot replace it.

Induction is exact:

$$\psi(y, \chi_q) - \psi(y, \chi_r) = - \sum_{\substack{p|q \\ p \nmid r}} \log p \sum_{p^j \leq y} \chi_r(p)^j.$$

Its absolute value is at most $u\omega(q)$. The principal character has the same lost-prime-power correction. Averaging over the actual characters does not add an extra $\varphi(q)$; summing over

$q \leq Y$ costs at most $2YL^2$. It is absorbed by $2NL^{-23/2}$. The printed inequality (12) in [8], which omits this logarithmic factor, is not used. Thus the full corrected mean satisfies

$$\sum_{q \leq Y} E^\#(q) \leq N[5252L^{-23/2} + \delta_{\text{low}}(N)], \quad (61)$$

where

$$\delta_{\text{low}}(N) = \frac{3L}{N} \sum_{2 \leq r \leq R} \frac{1}{\varphi(r)} \sum_{\chi \pmod{r}}^* \sup_{1 \leq y \leq N} |\psi(y, \chi) + \mathbf{1}_{\chi=\chi_*} H_{\beta_*}(y)|.$$

The suprema are required; an endpoint-only estimate does not prove this norm.

12.5 Reported primitive all-prefix certificate

The following certificate is the explicit analytic dependency of the low-conductor part. Its internal reviews and rational tests are recorded in the corpus. The proof below shows the reduction to direct remainder envelopes. It does not reproduce every lengthy remainder definition from [8]; the substitution asserted in part (b) is therefore identified as a reported project-derived analytic step, rather than as a new fully independently established theorem of this manuscript.

Lemma 12.1 (Reported primitive certificate). *Assume $u \geq 8 \cdot 10^6$, $R = \lfloor L^{16} \rfloor$, and select at most one primitive real Page pair up to R with*

$$\beta_* \geq 1 - \frac{1}{2(2.0452) \log R}.$$

Use the signed primitive explicit formula, the zero count and the local and common Page zero-free regions of [8]. Assume also the direct remainder substitution in part (b) below. Then, for each conductor $1 < r \leq R$,

$$\sum_{\chi \pmod{r}}^* \sup_{1 \leq y \leq N} |\psi(y, \chi) + \mathbf{1}_{\chi=\chi_*} H_{\beta_*}(y)| \leq 100N/u^{14},$$

and $\delta_{\text{low}}(N) < 40000\ell/u^{13}$.

Proof. (a) Fix the same height $T_0 = u^{34}$ for all characters and all prefixes. Since $R \leq 2u^{16} \leq u^{17}$, $\log(r(T_0 + 2)) \leq 52\ell$. The published local constant 6.3970 and common Page constant 2.0452 give, for every unselected zero in the signed sum with real part at least $1/2$,

$$1 - \Re \rho \geq \frac{1}{340\ell}.$$

The selected zero is removed before absolute values. Its reflected zero $1 - \beta_*$ remains in the remainder containing zeros with real part below $1/2$. The pair is selected once, not at a new height or modulus for each prefix.

For $y < N^{7/8}$, use $|\psi(y, \chi)| \leq \psi(y) \leq 2y$ and $H_{\beta_*}(y) \leq y - 1$. After at most $2u^{16}$ characters and division by N/u^{14} this costs at most $6u^{30}e^{-u/8}$. For $N^{7/8} \leq y \leq N$, the zero count is at most $200T_0\ell$ per character and $|\rho| \geq 1/2$. The remaining-zero sum, after the same normalization, is bounded by

$$800u^{65} \exp\{-7u/(2720\ell)\}. \quad (62)$$

The same lower value of $\log y$ applies to every character, so this allows the supremum inside each character sum.

(b) The direct remainder certificate uses the signed formula (21) and the direct remainders on pp. 1425–1429 of [8], with the half-integer Perron endpoint transported back to y . The asserted generous substitutions, uniform on the above long range, are

$$\begin{aligned} R_2 &\leq 1000Nu\ell/T_0, & R_3 &\leq 100Nu^2/T_0, \\ R_5 &\leq 36uT_0, & R_7 &\leq 100000N\ell^2/T_0, \\ R_8 &\leq 4000\ell/T_0, & R_{11} &\leq 10000\sqrt{N}u^{10}, \end{aligned}$$

with the additional $Nr_4/(T_0 - 1)$ at most $200N\ell/T_0$. For clarity, the substitution checks recorded in the project are: $r_1(r, 1) \leq 10\ell$, $r_1(r, T_0 \pm 1) \leq 16\ell$, $r_4 < 100\ell$, the shifted r_5 maximum below 250ℓ , and the shifted r_6 maximum below 70ℓ . In R_7 , the first factor is below $3N/T_0$ and the bracket below $11000\ell^2$. In R_8 , the first factor is below $12/T_0$ and the bracket below 270ℓ . In R_{11} , $\sqrt{r} \leq 2u^8$ bounds the first term by $18\sqrt{N}u^8\ell^2$, and the remaining bracket is below $1700\ell^2$ with multiplier below $2\sqrt{N}$. These estimates include the inverse gap of the reflected real zero. In R_3 the first $\log y$ term is additive, not a multiplier of the entire numerator. R_2 must keep both near-floor neighbors and the half-integer padding. These last substitutions are the reported analytic certificate, not a consequence of rational arithmetic alone.

After the characters and normalization, the complete direct fee is bounded by

$$\frac{2000\ell}{u^3} + \frac{200}{u^2} + \frac{200000\ell^2 + 400\ell}{u^4} + 20000u^{66}e^{-u} + 20000u^{40}e^{-u/2}. \quad (63)$$

Changing y^{β_*}/β_* to $H_{\beta_*}(y)$ adds the constant $1/\beta_* < 2$ on only the selected character; it is included in the exponential group. The source's Table 6 and equation (33) are not inputs of this certificate.

(c) At $u_0 = 8 \cdot 10^6$, $u_0 < 2^{23}$ and $15 < \ell_0 < 16$. The zero exponent in (62) exceeds 1280. Using $e > 8/3$ gives $800 \cdot 2^{1495} (3/8)^{1280} < 10^{-80}$. The first four terms of (63) total less than 10^{-6} , and its exponential terms and the short-prefix fee are less than 10^{-100} . Each term decreases thereafter; for the zero fee its logarithmic derivative in $\log u$ is at most $65 - (7/2720)(u/\ell)(1 - 1/\ell) < 0$. There is ample slack for the conservative bound $100N/u^{14}$. Finally,

$$\delta_{\text{low}} \leq 900L(1 + 17\ell)/u^{14} \leq 32400\ell/u^{13} < 40000\ell/u^{13}.$$

This proves the stated implications of the certificate. $\square$

The numerical direction of this bound matters. It is less than 10^{-70} , not 10^{70} . A stronger immediate consequence is

$$\delta_{\text{low}} < \frac{640000}{(8 \cdot 10^6)^{13}} < 2 \cdot 10^{-84} \quad (u \geq 8 \cdot 10^6), \quad (64)$$

because ℓ/u^{13} decreases and $\ell_0 < 16$. These are upper bounds for a corrected norm, not measured sizes of a Goldbach error.

Combining (60) and (61), two-prefix Cauchy gives the complete weighted remainder

$$\frac{|W^\#|}{N} \leq 24\sqrt{5252L^{-17/2} + L^3\delta_{\text{low}}}. \quad (65)$$

The factor 24 already contains both endpoints X and V . It is not doubled a second time. At u_0 , retain $L^{-17/2} < u_0^{-8}/2828$ and use $L^3\delta_{\text{low}} \leq 320000\ell_0/u_0^{10}$. The exact rational comparison

$$576 \left[\frac{5252}{2828u_0^8} + \frac{320000 \cdot 16}{u_0^{10}} \right] < \frac{64}{10^{54}}$$

therefore proves the reported sharpened allowance $|W^\#| < 8 \cdot 10^{-27}N$, conditional on Lemma 12.1. This is not the exceptional main term.

12.6 A zero-avoiding contour for the genuine Type-I head

For the adaptive quarter-profile evaluation assume now

$$u \geq 10^{24}, \quad U = V = 1, \quad \delta = \frac{1}{\sqrt{u}\ell^4}, \quad E = \delta u = \frac{\sqrt{u}}{\ell^4}, \quad T_1 = e^{E/120}, \quad h = T_1^{-1/2}, \quad J = 32. \quad (66)$$

Use the same Page pair up to R as before. In the Type-I head $d \leq u^8$, exclude only that one primitive character if its conductor $r > u\ell^4$. If it is absent, outside the head, even, non-square-free or nonunit, its actual head contribution is zero. All other characters on every multiple of its conductor are retained.

For a retained primitive nonprincipal character of conductor $q \leq u^8$, and any K with $\log K \leq 2u$, use the real arithmetic series

$$F(s) = \sum_{n \geq 1} \mu(n) \chi(n) \mathbf{1}_{(n,K)=1} n^{-s} = \frac{1}{L(s, \chi)} \prod_{p|K} (1 - \chi(p) p^{-s})^{-1}.$$

The series identity initially holds on $\Re s > 1$. The local published real-zero gap is $1 - \beta \geq 100/(\sqrt{q} \log^2 q)$ when a local Siegel zero exists. For the selected $r \leq u\ell^4$, this is at least 25δ . Every other local real zero has common Page gap above $1/(80\ell)$. The ordinary zero-free-region check is

$$28\delta(8\ell + E/120 + 1) < 1.$$

It follows that the disks of radius 4δ centered at $1 + 2\delta + it$, $|t| \leq T_1$, are zero-free. No possible reciprocal- L pole is crossed.

Here is the finite growing-mask bound. If $\delta \leq \min(1/4, 1/(2\ell))$ and $\Re s \geq 1 - \delta$, then

$$\left| \prod_{p|K} (1 - \chi(p) p^{-s})^{-1} \right| \leq u^6.$$

Indeed $p^{-1+\delta} < 3/5$ and $-\log(1-t) \leq (5/2)t$ on that range. For $p \leq u$ enlarge to the integers and use $p^\delta < 2$ to bound the sum by 2ℓ . There are at most $2u/\ell$ mask primes above u , with sum at most $4/\ell$. Thus the logarithm of the product is at most $5\ell + 10/\ell \leq 6\ell$. Every actual mask prime remains.

Nonprincipal periodicity supplies $|\sum_{n \leq t} \chi(n)| \leq q$. Partial summation continues L to $\Re s > 0$ and bounds it by $q|s|/\Re s$. On the larger disk use $|L(s)| \leq 4q(T_1 + 3)$. At its center the Euler logarithm is bounded by $\log \zeta(1 + 2\delta) \leq \log(1/\delta)$. Borel–Caratheodory with radii $4\delta, 3\delta$ then gives

$$|1/L(s, \chi)| \leq [4q(T_1 + 3)]^6 \delta^{-7}$$

on the inner disk. On the rectangle $1 - \delta \leq \Re s \leq c = 1 + 1/\log x$, $|\Im s| \leq T_1$, $N^{1/8} \leq x \leq N$, this proves

$$|F(s)| \leq 2^{24} u^{74} T_1^6. \quad (67)$$

Here $q^6 \leq u^{48}$, $\delta^{-7} = u^{7/2} \ell^{28}$ and $\ell^{28} \leq u^{14}$ give exponent $6 + 48 + 7/2 + 14 < 74$.

Let $M_f(x) = \sum_{n \leq x} \mu(n) \chi(n) \mathbf{1}_{(n,K)=1}$. Use the exact multiplicative box average

$$M_h(x) = h^{-32} \int_{[1, 1+h]^{32}} M_f(x t_1 \cdots t_{32}) dt_1 \cdots dt_{32}.$$

Its absolutely convergent Mellin representation is

$$M_h(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} F(s) x^s G_h(s) \frac{ds}{s}, \quad G_h(s) = \left[\frac{(1+h)^{s+1} - 1}{h(s+1)} \right]^{32}.$$

The apparent singularity at $s = -1$ is removable. To justify the identity, average the Mellin step kernel over the box. Its averaged weight is continuous, zero below $(1+h)^{-32}$ and one above 1; its Mellin transform is $G_h(s)/s$. The decay $|t|^{-32}$ gives absolute integrability and licenses Fubini on $\Re s = c > 1$. Also $|M_h - M_f| \leq 64hx + 1$. The $+1$ is the integer endpoint, not an omitted small error.

On the rectangle $|G_h| \leq 2$; for $|t| \geq T_1$, $|G_h(s)| \leq [3/(h|t|)]^{32}$. Truncating and shifting to $\Re s = 1 - \delta$ crosses neither a zero of L nor the pole at $s = 0$. Divide every resulting fee by x/u^4 . The five upper bounds are precisely

$$\begin{aligned} A_1 &= 2^{26}u^{79}e^{-3E/40}, & A_2 &= 6 \cdot 2^{24}3^{32}u^{78}e^{-11E/120}, \\ A_3 &= 6 \cdot 3^{32}u^5e^{-2E/15}, & A_4 &= 64u^4e^{-E/240}, & A_5 &= u^4e^{-u/8}. \end{aligned} \quad (68)$$

A_1 is the shifted vertical integral: $x^{-\delta} \leq e^{-E/8}$, the inverse norm costs T_1^6 , and the logarithmic integration costs at most u . A_2 is the horizontal integral, whose smoothing decay T_1^{-17} beats that T_1^6 . A_3 is the two tails on the original absolute-convergence line, using $|F(c+it)| \leq 2u$. A_4 and A_5 are the box-averaging fee and its endpoint. An unsmoothed degree-one Perron estimate would not justify A_2 .

At $u_0 = 10^{24}$, $u_0 < 2^{80}$, $54 < \ell_0 < 56$ and $E_0 > 10^{12}/56^4 > 100000$. With $e > 2$, the five fees are below

$$2^{6346-7500}, \quad 2^{6331-9166}, \quad 2^{467-13333}, \quad 2^{326-416}, \quad 2^{320-10000},$$

respectively, hence each is below $1/8$. Their uniform monotonicity follows from $d \log E / d \log u = 1/2 - 4/\ell > 2/5$ and $a - (2/5)bE < 0$ for each u^ae^{-bE} above. We obtain the explicit retained nonprincipal bound

$$|M_f(x)| < x/u^4 \quad (N^{1/8} \leq x \leq N). \quad (69)$$

This proof controls $1/L$ itself and avoids the possible pole. It does not infer a Mertens-in-progressions estimate from a ψ -formula, and does not assume a bound for $1/L'(\beta_*)$.

12.7 Whole Type-I error accounting, including the paid primitive line

The genuine scalar is $I = I_D - I_W$. A complementary-divisor switch, with the strict face retained, gives

$$I_D = - \sum_{\substack{k \leq Q \\ (k,N)=1}} \mu(k)^2 \sum_{\substack{\alpha < r \leq \lfloor (N-1)/k \rfloor \\ (r,kN)=1}} \mu(r) \log r \log(N - kr).$$

The identity used is $\mu(k)\mu(kr) = \mu(k)^2\mu(r)\mathbf{1}_{(k,r)=1}$. For $K \leq N^2$, ordinary masked Mertens convolution split at $\sqrt{x}$ gives, for $x \geq N^{1/4}$,

$$|M_K(x)|/x \leq 26u^2e^{-\sqrt{u/48}} + e^{-u/32}.$$

The mask-smooth coefficient has reciprocal moment at most $2(1 + \log K)$ and half-moment at most $e^{12\sqrt{u}}$. The weight $\log t \log(N - kt)$ has supremum at most u^2 and variation at most $2u^2$. Abel summation, with both endpoints, then costs at most $4Ru^2$ times the normalized prefix bound. The outer sum is harmonic, not of size Q . Thus

$$|I_D|/N \leq 208u^5e^{-\sqrt{u/48}} + 8u^3e^{-u/32}.$$

Let

$$T_{\text{sing}} = \sum_{\substack{1 \leq n < N \\ (n,N)=1}} \mu(N - n) \log n \mathfrak{S}(nN).$$

For physical $m = N - n \geq \sqrt{N}$, the exact prefix $\min(Q, \lfloor (m-1)/\alpha \rfloor)$ is at least $N^{1/5}$. Its mask is nN , and (53) has main $-\mathfrak{S}(nN)$, not $-\mathfrak{S}(N)$ on a general log-weighted row. All physical short m , including empty prefixes, cost at most $8\sqrt{N}L^3$. The literal completion therefore gives

$$|I - T_{\text{sing}}|/N \leq 4 \cdot 10^8 u^5 e^{-\sqrt{u/60}} + 160u^2 e^{-u/40} + 8L^3 e^{-u/2} \\ + 208u^5 e^{-\sqrt{u/48}} + 8u^3 e^{-u/32} < u^{-4}$$

at the adaptive onset, by elementary decreasing-envelope comparisons.

Define the positive odd-square-free coefficient $g(d) = \prod_{p|d} (p-2)^{-1}$, $g(1) = 1$, zero off that support. Finite Euler expansion gives exactly

$$\mathfrak{S}(nN) = \mathfrak{S}(N) \sum_{\substack{d|n \\ (d,N)=1}} g(d),$$

hence T_{sing} is a weighted sum of actual reduced classes $m \equiv N \pmod{d}$, with the unit- N mask retained. Its coefficient moments are

$$\sum_{d \leq D} g(d) < 8(1 + \log D), \quad \sum_{d > D} \frac{g(d)}{d} < \frac{6(1 + \log D)}{D} \quad (D \geq e).$$

For the second bound, compare local Euler factors to obtain $\sum g(d)d^{-\sigma} < 2\zeta(1+\sigma) \leq 2(1+1/\sigma)$. For $\sigma = 1/\log D \leq 1$, Rankin's inequality gives the displayed tail. For the first, factor $g = h * (1/n)$; the absolute moment is $(3/2) \prod_{p>2} (1 + 3/[p(p-2)]) < 8$, by telescoping the odd reciprocal quadratic series.

Take $D = u^8$. Orthogonality on each actual d induces a primitive character with exact mask $K = Nd$, $\log K \leq 2u$. There is no unrestricted-conductor dimension factor: its $\varphi(d)$ characters cancel its own $1/\varphi(d)$. The principal character is separate. Ordinary masked Mertens, now calibrated at $x \geq N^{1/8}$, bounds its normalized prefix by $13u^2 e^{-\sqrt{u/96}} + e^{-u/64} < u^{-4}$. For smaller prefixes all characters use the direct maximum $N^{1/8} < N/u^4$. Abel summation for the decreasing weight $\log(N-m)$ costs at most u . The retained head, bridge and entire $d > u^8$ tail, normalized by $N/(u\ell)$, total at most

$$\frac{\ell}{u^3} + \frac{648\ell(1+\ell)(1+8\ell)}{u^2} + \frac{486\ell(1+\ell)(1+8\ell)}{u^6} < 10^{-12}. \quad (70)$$

The absence of a $+1$ in this positive divisor tail is justified by the original positive multiples $n = dj$, whose count is exactly $\lfloor (N-1)/d \rfloor \leq N/d$. It is not an arbitrary interval count.

The one excluded primitive character is paid on all its actual multiples. Its positive coefficient mass is

$$P_g(r) = \sum_{r|d} \frac{g(d)}{\varphi(d)} < 3 \frac{g(r)}{\varphi(r)} < \frac{6}{r^2} \left(\frac{r}{\varphi(r)} \right)^3 < \frac{162(\log \log r)^3}{r^2}.$$

The first Euler product is below $e < 3$; the second comparison uses $\prod_{p|r} (1 + 1/[p(p-2)]) < e^{1/2} < 2$. The last step uses the universal Rosser–Schoenfeld totient bound [15]; its 2.50637 version covers its small exceptional integer. Also $\mathfrak{S}(N) < 3\ell$. Exactly one induced character occurs on each multiple, not $\varphi(d/r)$ characters. Bounding the genuine masked, log-weighted inner sum by Nu pays this line with normalized allowance

$$486 \frac{[\log(\ell + 4 \log \ell)]^3}{\ell^6} \leq 486 \frac{[\log(2\ell)]^3}{\ell^6} < \frac{486 \cdot 125}{54^6} < \frac{1}{400000}.$$

This decreases with ℓ . Together with (70), it proves the reported $|I| < 3 \cdot 10^{-6} N/(u\ell)$. No active modulus has been discarded, and no favorable character phase has been assumed.

12.8 Whole E_J , centering corrections and exceptional main

The literal omitted-divisor union reduces on this cap family to $J(m) = -\sum_{d|m, d \leq \alpha} \mu(d) \log d$. Indeed $Q \geq \alpha$ and $Q(\alpha+1) > X$, so the other union branch cannot occur for $d > \alpha$. Expanding $\mu(m)^2$ before grouping gives the exact error allocation

$$E_J = R_{\text{main}} + \sum_q w_B(q) \Delta^*(q) - T_B,$$

where

$$\Delta^*(q) = \psi(X; q, N) - \psi(V; q, N) - U_N(q; V, X) - \frac{X - V}{\varphi(q)}.$$

U_N is the exact lost nonunit prime-power mass. T_B is the signed physical remainder with $B < b \leq \sqrt{X}$ in the square-free expansion; it is not the infinite model tail used above. The elementary AP count $\psi_s(X; q, N) \leq (N/q + 1)u$ and the $b = gh, d = gk$ decomposition give

$$|T_B|/N \leq 2u^2 L(2 + \log B)/B + 2u^2 e^{-u/4} \leq 46L^{-16} + 2u^2 e^{-u/4}.$$

The second term pays the physical counting $+1$ and must remain.

Put $\Omega = \psi(X) - \psi(V)$ and $J_\beta = (X^{\beta*} - V^{\beta*})/\beta_* \leq N$. On every active modulus, including multiples of r_* , the two-prefix identity is

$$\sum_q w_B(q) \left[\psi(X; q, N) - \psi(V; q, N) - \frac{\Omega}{\varphi(q)} \right] = -\chi_*(N) J_\beta M_{r_*}(B) + W^\#.$$

Inserting (56), the exceptional main is

$$+\chi_*(N) \mathfrak{S}(N) \kappa(r_*) J_\beta.$$

Its effective sign depends on $\chi_*(N) \mu(r_*)$. It is not automatically favorable. The projection error and $W^\#$ have different error components.

Recentring from Ω to $X - V$ adds exactly $(\Omega - (X - V))/\varphi(q)$, and the retained unit correction subtracts $U_N(q; V, X)$. The exact endpoint identity is

$$\Omega - (X - V) = (\psi(N) - N) - \Lambda(N) + 1 - (\psi(V) - V).$$

It retains $\Lambda(N)$, including $N = 2^t$. From (58) and $\sum 1/\varphi(q) < 3L$, $\sum |w_B(q)|/\varphi(q) < 8L^2$. The ordinary published estimate in [9] implies the safe weakening $|\psi(N) - N| \leq 34Nu^2 e^{-(4/5)\sqrt{u}}$. The endpoint and lower-prefix cost is at most $8L^2(u + 1 + 3V)$. The nonunit prime-power mass is at most $2u^2$, and $\sum |w_B(q)| \leq u\alpha B(1 + \log B)$. Their joint variable envelope is therefore

$$272u^2 L^2 e^{-(4/5)\sqrt{u}} + 220L^{24} e^{-3u/4}$$

after division by N . At $u \geq 8 \cdot 10^{10}$ this is at most $2u^{-4}$. All the projection, model and physical-tail rates then give, using (65), the total nonexceptional allowance

$$|\text{nonexceptional components}|/N \leq 14411u^{-4}, \quad 14411 = 14400 + 3 + 4 + 2 + 2.$$

The first constant pays $W^\#$ by $24\sqrt{(5252 + 320000)/u^8} < 14400/u^4$; the next constants pay projection, R_{main} , T_B and the corrections exactly once.

The exceptional main requires two alternative cases, not two added fees. If $r_* > u^{3/2}$, the totient bound and $\prod_{p|r} (1 + 1/[p(p-2)]) < 2$ give its normalized cost

$$54\ell^2 [\log(1.5\ell)]^2 / \sqrt{u}.$$

If $3 \leq r_* \leq u^{3/2}$, the published local gap and the coefficient bound $\mathfrak{S}(N)|\kappa(r)| \leq 54\ell^3/r$ give at most $2 \cdot 54\ell^3 r^{-1} \exp\{-100u/(\sqrt{r} \log^2 r)\}$ before normalization. Using $\log r \leq 1.5\ell$ and setting $t = 1/\sqrt{r}$, the elementary maximum of $t^2 e^{-at}$, namely $4e^{-2}/a^2$, yields

$$|\text{small-conductor main}|/N \leq (2187/10000)\ell^7/u^2.$$

At $u_0 = 10^{24}$ the normalized large and small alternatives are respectively below $54 \cdot 56^2 \cdot 25/10^{12} < 9/(2 \cdot 10^6)$ and $0.2187 \cdot 56^8/10^{24} < 10^{-10}$. Both functions decrease thereafter. Adding the nonexceptional $14411\ell/u^3 < 10^{-12}$ gives the reported whole bound

$$|E_J| < 5 \cdot 10^{-6} \frac{N}{u\ell}.$$

This conclusion depends on the all-prefix certificate, not just a small endpoint value or an exceptional-modulus-free theorem.

12.9 The complete centering component and detector losses

Use the literal endpoint $x(m) = \min(Q, \lfloor (m-1)/\alpha \rfloor)$ in $W_Q(n, m)$. For a retained prime row $n = p > Q$, all $k \leq x(m)$ are less than p , so the finite mask nN is exactly the mask N before taking any limit. On the further physical bulk $m > \lceil \sqrt{N} \rceil$, $x(m) \geq N^{1/5}$; (54) therefore gives main $-\mathfrak{S}(N)$. Its total Mangoldt weight is at most Nu , so its whole error is bounded by N times that envelope.

The literal physical partition keeps low primes $p \leq Q$, primes with short m , proper prime powers, $n \leq V$ omissions and nonunit omissions. The separate original corner estimates sum to

$$6QL^3 + 6\lceil \sqrt{N} \rceil L^3 + 12\sqrt{N}L^4 + 3VL^2 + 6L^3 \leq 40N^{3/4}L^4.$$

This is the reported literal corner certificate; it does not apply the bulk mask to those other rows. Together with the bulk it gives the complete, variable error envelope

$$|E_Z|/N \leq 4 \cdot 10^8 u^5 e^{-\sqrt{u/60}} + 160u^2 e^{-u/40} + 40L^4 e^{-u/4}. \quad (71)$$

The proper powers here belong to the one-Mangoldt centering component, not to the two-Mangoldt detector conversion. The latter has its own nonnegative retained proper-power mass, satisfying

$$P_{\text{retained}} \leq \sqrt{N}(\log N)^2.$$

For each prime base $p \leq \sqrt{N}$, the sum of $\log p$ over its proper powers below N is at most u , and the second Mangoldt factor is at most u , proving this estimate.

Normalize (71) and the detector estimate by $N/(u\ell)$. At $u_0 = 10^{24}$ every negative exponent is larger than 10^{11} , while the positive factors are below $2^{516}, 2^{254}, 2^{416}$ and 2^{246} , respectively. They are individually below 2^{-1000} ; the corresponding polynomial-exponential envelopes decrease. Thus

$$|E_Z| < 10^{-12}N/(u\ell), \quad P_{\text{retained}} < 10^{-12}N/(u\ell).$$

The omitted-prime-pair selector loss T_{pairs} is nonnegative and appears with a favorable minus sign. It need not be charged as an additional positive upper error.

12.10 Conditional exterior conclusion and the unresolved target

Under the analytic certificate and literal completion estimates specified above, the quarter-power assembly gives

$$\begin{aligned} |I| &< 3 \cdot 10^{-6}N/(u\ell), & |E_J| &< 5 \cdot 10^{-6}N/(u\ell), & u &\geq 10^{24}, & U = V = 1. \\ |E_Z| &< 10^{-12}N/(u\ell), & P_{\text{retained}} &< 10^{-12}N/(u\ell), \end{aligned} \quad (72)$$

Their useful exterior sum is

$$\epsilon_{\text{tr}} = P_{\text{retained}} - T_{\text{pairs}} + E_J + E_Z - I < (8 \cdot 10^{-6} + 2 \cdot 10^{-12})N/(u\ell).$$

This reevaluates the original components; no old fixed allowance is added to it. The universal reference lower bound from the separate prime-square argument is $\mathfrak{S}(N)(N - F_N) \geq 2541N/(524288u\ell)$. If the exact covered/uncovered bridge holds and if the still-open analytic target

$$D_N \leq N/(256u\ell)$$

is established on the same support and profile, the terminal ledger would leave the positive coefficient

$$\frac{2541}{524288} - \frac{1}{256} - \frac{8}{10^6} - \frac{2}{10^{12}} = \frac{119337327869}{128000000000000} > 0.$$

The equality of rational numbers is a certificate of a conditional margin. Neither the covered bridge nor the required bound for D_N is proved by this arithmetic. A smaller Type-I component also does not make D_N smaller: at fixed V, α, Q , changing U redistributes the same arithmetic term between I_U and D_U , with its covered positive error still present.

12.11 Relation to prior work and responsible scope claims

The external tools are classical: ordinary Mertens cancellation, primitive character mean values, zero-free regions, partial summation, Euler products, Borel–Caratheodory and a smoothed Perron contour. Their coefficient-specific combination and explicit error allocation are project deductions; no priority claim for a new distribution theorem has been established. The formal contribution concerns exact finite bridges and certificates with declared hypotheses. It does not establish binary Goldbach, overcome the parity barrier or prove an analytic infinite-range statement in Lean.

Sedunova’s two papers have different bibliographic identities. The 2018 Besancon paper is *A partial Bombieri–Vinogradov theorem with explicit constants*; the 2019 Bordeaux paper [12] is *A logarithmic improvement in the Bombieri–Vinogradov theorem*, 31(3), 635–651. A corollary with an effective implied constant does not provide a particular numerical constant unless it is evaluated. The explicit assembly here uses the stated AH primitive theorem, not an invented numerical Sedunova constant.

M. Ram Murty and A. Vatwani, *A remark on a conjecture of Chowla*, Journal of the Ramanujan Mathematical Society 33(2) (2018), 111–123, is relevant qualitative context for raw Mertens cancellation in progressions with maximal prefixes. Such a theorem cannot by itself provide the numerical 10^{24} budget here; the manuscript’s explicit route treats the reciprocal- L function and its mask directly. The latest author publication page confirms the 2018 journal entry, but the former institutional PDF redirects. No fresh primary-PDF verification of its Proposition 3 is claimed in this manuscript support review.

Weighted sieves with switching [13] and the sieve framework [14] support the separate qualitative sector. That sector has a different α and an unevaluated onset. In particular, the printed sign of the absolute remainder in Richert’s (11.80) is not used as a lower-bound input; the adopted sector proof uses a good/bad partition and the stated modern linear-sieve lemma instead. Chen-type prime plus at-most-two-prime-factor conclusions do not imply a prime plus prime conclusion or a sign comparison of proper semiprime and triple-prime masses. Prime credit must remain in the exact terminal row.

The historical repository’s own `AUDIT_NOTE.md` states that four interface predicates were weak enough to admit trivial witnesses, while `PCBASymptotic` already expresses asymptotic Goldbach. Accordingly, compiling a conditional interface theorem is not a proof of the corresponding number-theoretic hypothesis. The present corpus should be described by its exact

theorem statements, source hashes, axiom checks and open assumptions, without an unsupported claim of being a first formal proof of Goldbach.

Finally, arXiv’s published policy [21] requires disclosure of significant generative-tool use and keeps responsibility for all content with the named human authors; a language model is not an author. An appropriate disclosure is that generative tools assisted research, proof drafting, code generation and language editing, while the human author takes responsibility for the mathematical statements and references. The reviews in this corpus are internal computational-assistant reviews. An arXiv-friendly source package is a technical format, not a guarantee of moderation acceptance or external peer review.

13 The unresolved terminal estimate

The reference lower bound and a numerical loss allowance must be compared on the same support and the same parameter profile. In the quarter profile, a sufficient covered-route target remains

$$D_N \leq \frac{N}{256 \log N \log \log N}. \quad (73)$$

It is not an established theorem, and it is not asserted necessary. If the reported primitive input, the exact covered identity and the stated written exterior assembly apply, (73) leaves the conditional coefficient

$$\frac{2541}{524288} - \frac{1}{256} - \frac{8}{10^6} - \frac{2}{10^{12}} = \frac{119337327869}{128000000000000} > \frac{9}{10000}.$$

The positive rational comparison is a consequence of the premises. It does not discharge (73), nor prove that the covered decomposition has the required complete support.

The complete direct scalar formulation avoids the artificial covered excess:

$$R_{sf} = \mathfrak{S}(N)(N - F_N) - F_{\text{fixed}} - P_{\text{retained}} + T_{\text{pairs}} - E_J - E_Z.$$

An upper bound for $F_{\text{fixed}} + E_J + E_Z + P_{\text{retained}} - T_{\text{pairs}}$ below the reference would suffice in that convention. It must not be added to the covered-route losses as if they were different errors. The cutoff-invariance identity in the formal appendix shows that completing a Vaughan fibre redistributes the same prime row between the short and long components. It does not supply new cancellation simply by changing U .

The eighth-profile sector result has a different remaining object:

$$F_{\text{fixed}} = F_C + F_{\text{rough}} + F_{\text{nonrough,remaining}}.$$

The two first supports are disjoint and have a negative combined upper bound. The third includes potentially large positive rows; no usable global upper bound has been obtained for it. In addition, the quarter-profile effective exterior bounds have not been transferred to this eighth profile. A negative coefficient already included in J_{ref} cannot also be spent as an extra credit after the whole J_{ref} has been replaced by its main term.

13.1 What a subsequent proof would have to add

A future argument must supply a quantitative estimate for the actual signed aggregate, with its masks, weights, strict faces and complement retained. It may use a combined estimate rather than separate absolute bounds, but no proof has shown that every possible absolute or spectral method fails. The $N/64$ obstruction concerns the particular positive envelope tested. Similarly, a large Gram eigenvalue rules out a specified uniform operator bound; it does not

determine the projection of the actual arithmetic vector. A failed local sign matching shows that matching is not automatic, whereas other complete fibres do cancel at one fixed CRT root. For coprime a, r , that root is unique modulo ar ; multiple sign conductors inside the same ar are a different matter.

Once a sufficient large- N estimate is proved, its constants and onset must be effective and compatible with every external input. Finally the entire interval below that onset must be covered by an independent theorem or verified computation. A statement above $N \geq \exp(10^{24})$ would leave an enormous gap above the published finite verification range. The present report supplies no coverage of that gap.

No extension to twin primes or to other additive conjectures is implied. The shared vocabulary of Type-II decomposition and sieve masks is not itself a transfer theorem. For one fixed N there are only finitely many prime representations; any prospective quantitative conclusion concerns their finite number as N varies.

14 Corpus audit, reproducibility and documentary scope

The synthesis uses the supplied local mathematical collections, the accessible Google Drive research folder and the public GitHub repository. Corpus triage and file integrity are distinct from proof validation. The electronic coverage record lists inspected sources and the exclusions; no whole-computer proof review is claimed.

Collection	Inventory evidence	Review scope
Local earlier manuscripts	936 selected files; 616 PDF copies; 560 distinct PDF hashes; 5,643 extracted pages	Retained extraction index from the initial audit; selected mathematical chains read in detail
Current formal corpus	894 physical Lean files; 502 distinct hashes; 71 canonical recent sources	Deduplication, theorem headers, source-bound logs, receipts and axiom vocabulary checked
Google Drive	900 folders visited; 770 PDF metadata entries outside dependency/build/Git exclusions	Recursive research metadata and selected source text; pending dependency directories excluded, not silently treated as reviewed proofs
GitHub	1,254 tree nodes; response not truncated	Pinned tree and interface audit read; conditional historical architecture separated from recent certificates
Primary literature	Versioned arXiv articles, publisher and author-hosted sources	Theorem applicability and proof inputs identified; no exhaustive literature novelty certification

Table 4: Coverage evidence. Metadata counts include copies and are not counts of independently proved mathematical results.

The GitHub snapshot is commit

979c05416140504f11c6e9ba6ad1dd565838bea1.

It was retrieved from <https://github.com/seisner1967-hash/goldbach-horizon>. The supplied Drive entry is the Project Horizon research folder. The recursive inventory stopped traversing third-party Mathlib/build/Git directories after classifying the remaining queued directories; it also excluded unrelated topics. Source documents such as CS21 and the E4 family

were retrieved directly. The descriptive *L’Horloge Invisible* is background context rather than an arithmetic proof. Retrieved documents about other conjectures are outside this report’s mathematical scope.

14.1 The historical interface audit

The repository’s own `AUDIT_NOTE.md` records that four of five old interface predicates were trivially satisfiable or not sufficiently connected to the intended arithmetic operators. Its `PCBASymptotic` predicate already asserts eventual positivity of the Goldbach representation count for every even integer. Consequently inhabiting that architecture cannot be described as an independent proof of asymptotic Goldbach. Historical source flags include gaps, custom axioms and finite native certificates; these are separated from the canonical recent standard-axiom inventory. The mathematical content of an individual finite certificate remains meaningful within its exact scope, but a repository-wide assertion of “no gaps and no nonstandard axioms” would be false.

14.2 Frozen packages and current checks

The three immutable packages preserve complementary evidence. Their complete SHA256 values are in the electronic manifest; Table 5 gives distinguishing prefixes. During final preparation all ZIP CRC, payload byte sizes and manifest hashes were replayed: 398/398, 67/67 and 7/7 payloads matched. This final preparation did not recompile Lean or re-execute every previously reported finite test.

Package	Payloads	New latest conclusions	SHA256 prefix
Sprint21 audit snapshot	398	16	3b88e4f9367fb2b9
Reviewed continuation addendum	67	3	66dc5eecb878c78a2
Post-freeze rough supplement	7	0	9d44c84e76e78af0

Table 5: Frozen evidence packages. Manifests are not counted as payloads. The rough supplement does not add Lean theorems.

The companion folder contains the standalone English TeX source, the full electronic formal-statement registry, 71 canonical Lean sources, original compilation scripts, retained receipts and axiom logs, selected written proofs and exact-constant checkers. Lean4.15.0 and the exact Mathlib revision are pinned. A portable replay driver accepts an explicit compiler, the pinned dependency library directories and a fresh build directory; it performs no installation or network update. The original scripts retain their host-specific provenance. An old matching compiled file is not a fresh source replay.

The computational examples are reproducible with the retained Python scripts. The initial grid checker reports 51,000 defect cases and 3,072 covariance cases; subsequent reports include exact rational onset guards and finite fibre tests. These counts describe the individual delivered checkers and are not added into an artificial total of independent mathematical evidence. The complete axiom-audited theorem appendix is supplied electronically instead of printing hundreds of historical helper statements inside the main analytic narrative.

14.3 Public, versioned reproduction material

The frozen evidence snapshot is publicly available at the following commit-addressed directory:

Project Horizon: reproducibility snapshot, 1 October 2026.

The exact revision is

8ef89b3cef4cddffd9bf9ca81141280b31ea6af6.

The file `goldbach_verification_bundle.zip` contains 362 manifest-listed payloads. Its SHA256 is

310c63cdb09e9a190da5bc01ddd781515a7cc717d73b680c8f30ef8fae7b8625.

This is the unchanged evidence snapshot accompanying version 1. The present editorial revision changes presentation and scope explanations, without rewriting retained Lean sources or historical receipts. Within the archive, `review/lean_inventory_full_statement_appendix.txt` supplies the complete statement registry; `reproducibility/lean` supplies canonical sources, dependency pins and replay instructions. The selected analytic notes retain the unresolved primitive-input boundary. Historical wording in archived notes is provenance; the current manuscript governs current claims.

The manuscript and its standalone TeX source are CC BY 4.0. This licence does not relicense software dependencies or historical evidence files. No DOI, journal acceptance, or fresh Lean compilation is inferred from this public repository link.

14.4 AI assistance and author responsibility

Substantial generative-AI assistance was used in corpus retrieval, code generation, derivation, drafting and internal cross-checks. The named human author is Durand Serge; no AI system is listed as an author. This version has internal source and numerical checks, but has not undergone external mathematical peer review. The human author must inspect and accept responsibility for its content before submission. This disclosure follows the arXiv policy on substantial AI assistance [21]. This public research report is formatted for arXiv-compatible source processing. Repository dissemination does not imply endorsement, moderation approval, journal acceptance or external mathematical validation.

15 Priorities for independent review

The next work should proceed in the following order, with each result retaining its evidence status.

1. **Complete the primitive analytic input.** For every remainder in the Bordignon-based argument, state the original formula, its domain, each substitution, and the final numerical majorant. Until this chain has been independently checked, Lemma 12.1 and its dependent estimates keep their conditional status.
2. **Estimate the actual signed complement.** Obtain a quantitative upper bound on the nonrough remaining sector or on the complete covered-route aggregate, with the original masks, logarithmic weights, endpoints and favourable terms retained. A new identity or a finer absolute envelope is not a substitute for this estimate.
3. **Reconcile parameters and thresholds.** Establish the complete transfer identity on the same support as the chosen analytic bounds. Do not import the effective quarter-profile allowance into the eighth profile without a new argument; then evaluate the resulting common onset.
4. **Reproduce and review.** Rebuild the canonical Lean sources with pinned dependencies, inspect the full declaration registry, rerun the rational checkers and obtain external specialist review of the analytic passages. Finally, any Goldbach theorem would require coverage of the entire interval below its effective onset.

This version is intended to make these tasks inspectable. It does not assert proximity to a proof or replace them by a publication claim.

A Finite Lean certificates, inventory and trust boundary

The printed listings render Lean Unicode symbols with ordinary mathematical fonts. The electronic statement inventory and bundled Lean sources retain their original Unicode text.

A.1 What the formal count means

The canonical numbered-sprint inventory contains 71 source files, 66 of which contain theorem or lemma declarations. There are 852 lexical declarations (833 public declarations and 19 private helpers). The individually printed, standard-axiom subset contains *658 distinct fully qualified name/statement pairs*; copies, extracted archives and repeated imports do not increase that count. These are accumulated finite identities, inequalities and conditional interfaces, not 658 estimates resolving the remaining arithmetic problem. The last research cycle contributes exactly *19 new public conclusions* in six modules: 16 in the immutable snapshot and three in its separate continuation addendum. Private helpers and old dependencies recompiled for a fresh replay are excluded from those 19.

Table 6: Individually axiom-audited formal families.

Phase	Count	Finite formal content and boundary
3	52	Fourier/Parseval and convolution; interval semantics; a finite spectral example and arithmetic variance/Gram identities.

Phase	Count	Finite formal content and boundary
4	132	Proxy coefficient energy, four-modulus combinatorics, anchors, diagonal and finite mass/simplex geometry; no signed cancellation.
5	51	Ramanujan/Fourier formulas, folded windows, smoothing and signed cofactor kernels; no final arithmetic contraction.
6	57	Exact Vaughan and finite Heath–Brown-type identities; Type-II rectangle assembly and variance reduction.
7–9	25	Four-sign structure, genuine beta-weighted fibers and literal cutoff/calibration geometry.
10–12	123	CRT/coprimalty masks, character/nonunit splitting, Kloosterman energy, conductor support and shifted dispersion sectors.
13–15	135	Multiplicity, diagonal and conditional off-diagonal budgets; finite Euler factors, margin interfaces and cofactor boundaries.
16–18	58	Ratio pushforward, finite Gauss factorization, Rosser boundary, signed assembly and actual parity/reference interfaces.
19	0	Explicit analytic prefix/mask transport was written outside Lean.
20	6	Actual Möbius-log fibers and exact truncated-prime branches.
21	19	The latest coefficient/support identities displayed below.
Total	658	Repeated imports and physical replay copies are not counted.

The accepted recent toolchain is Lean 4.15.0 (commit 11651562caae) with Mathlib pinned to 9837ca9d65d9de6fad1ef4381750ca688774e608. Every one of the latest 19 conclusions has a retained production compilation and a separate fresh replay, with exit code zero and source/log/compiled-artifact hashes. The printed axioms are drawn only from `propext`, `Classical.choice`, and `Quot.sound`. No latest theorem uses `sorry`, `admit`, `native_decide`, or a custom analytic axiom. The final inventory checked existing evidence; it did not run another compilation or turn written analytic estimates into Lean theorems.

The electronic `lean_inventory.json` records all canonical source hashes, exact headers and hypotheses, receipt and log hashes, imports, compile-script references and physical-copy groups. The full electronic statement appendix lists the 658 explicitly audited statements. The historical Horizon repository is separate: its own interface audit identifies weak predicates, while `PCBAsymptotic` already assumes eventual Goldbach positivity. Historical files containing gaps or custom axioms are not included in the latest trust claim.

A.2 Definitions needed to read the latest statements

Multiplication of arithmetic functions denotes Dirichlet convolution, $(f * g)(n) = \sum_{ab=n} f(a)g(b)$, and arithmetic functions vanish at zero. Write μ for the actual Möbius function, Λ for von Mangoldt, and φ for Euler’s totient. Lean’s natural subtraction is truncated subtraction and its divisions between natural numbers are integer divisions. `Real.log` is the total real logarithm; in particular it is defined at zero. No positive logarithm argument may be inferred without the displayed hypotheses. `n.divisors` and `n.divisorsAntidiagonal` are finite divisor sets; the latter consists of pairs whose product is n .

Local Euler projection and literal-cut coefficient. For a Boolean selector ε , put $d = p$ if it is true and $d = 1$ otherwise. The definitions in the first two modules are

$$\begin{aligned}\text{localProjection}(p, \varepsilon) &= \sum_{b \in \{1, p\}} \frac{\mu(b)\mu(d)}{\varphi(\text{lcm}(b^2, d))}, \\ w_\alpha(r, s) &= \mu(s)\mu(r) \sum_{t|s} \mathbf{1}_{rt \leq \alpha} \mu(t) \log(rt).\end{aligned}$$

The Lean name for the second definition is `signedDivisorCoefficient r s alpha`. The factor `s.divisors.card` in its bound is the actual number of divisors of s . The bound does not assume that the complete fiber is admitted.

Literal divisor and harmonic kernels. For $m = N - n$, define

$$\begin{aligned}D_{\alpha, Q}(m) &= \sum_{k|m} \mathbf{1}_{k \leq Q} \mathbf{1}_{\alpha k < m} \mu(k) \log(k/m), \\ W_{N, \alpha, Q}(n) &= \sum_{1 \leq k \leq Q} \mathbf{1}_{(k, N)=1} \mathbf{1}_{(n, k)=1} \mathbf{1}_{\alpha k < m} \frac{\mu(k)}{\varphi(k)} \log(k/m), \\ K_{N, \alpha, Q}(n) &= \mathbf{1}_{(n, N)=1} \mu(m) (D_{\alpha, Q}(m) - W_{N, \alpha, Q}(n)).\end{aligned}$$

`divisorKernel alpha Q m` is $D_{\alpha, Q}(m)$. `literalKernel N alpha Q n` is $K_{N, \alpha, Q}(n)$. The strict moving face is part of these definitions.

Let $\mu_{\leq U} = \mu \mathbf{1}_{n \leq U}$, $\mu_{> U} = \mu \mathbf{1}_{n > U}$, and $\beta_V = \Lambda \mathbf{1}_{n > V} * \zeta$, where the arithmetic function ζ equals one on positive integers. The code defines $T_{U, V} = \mu_{> U} * \beta_V$ and $A_{U, V} = \mu_{\leq U} * \beta_V$. Its three rows and its formal deficit are

$$\begin{aligned}\text{fullRow} &= \sum_{0 \leq n < N} T_{U, V}(n) K_{N, \alpha, Q}(n), \\ \text{smallRow} &= \sum_{0 \leq n < N} A_{U, V}(n) K_{N, \alpha, Q}(n), \\ \text{primeRow} &= \sum_{0 \leq n < N} \Lambda(n) \mathbf{1}_{n > V} K_{N, \alpha, Q}(n), \\ \text{ledgerDeficit} &= -\text{fullRow} + e + |e|.\end{aligned}$$

The name `primeRow` does not exclude proper prime powers of the first axis. The formal `ledgerDeficit` equals a proposed external ledger quantity only after its exact covered bridge is supplied; the module does not prove that bridge or a deficit bound.

Complete H and small-divisor credit. For natural y , put $\mu_{> y}(n) = \mu(n) \mathbf{1}_{n > y}$ and define the actual integer coefficient

$$H_y(n) = (\mu_{> y} * \mu_{> y} * \zeta)(n) = \sum_{d|n} \sum_{uv=d} \mu(u) \mathbf{1}_{u > y} \mu(v) \mathbf{1}_{v > y}.$$

The predicate `rough y n` used in this module means exactly $\forall d \mid n, d \leq y \Rightarrow d = 1$. The hypotheses explicitly require positive arguments where necessary; the special value $n = 1$ is retained in the rough formula. The final coefficient definition is

$$j_\alpha(m) = - \sum_{d|m, d \leq \alpha} \mu(d) \log d, \quad \text{smallDivisorCredit alpha m} = j_\alpha(m).$$

`IsPrimePow P` means that P is a positive prime power. The last theorem assumes both its negation and $\mu(P) = -1$; those assumptions are not conclusions of a density estimate.

A.3 The 19 exact latest-cycle declarations

The displayed headers retain every hypothesis and the original Lean parameter order. They omit only the proof bodies and normalize blank lines. Names are relative to the indicated namespace; the full qualified name is the namespace followed by the displayed theorem name. The lettered labels below are exposition labels, not additional Lean declarations. The subsequent analytic use of an identity is separate from its finite kernel-checked statement.

A.3.1 GoldbachActualEulerProjection

Namespace: `GoldbachResearch.ActualEulerProjection`.

This is a finite two-choice projection with actual Möbius signs and the actual lcm totient. No infinite Euler product or analytic remainder is formalized.

```
theorem actual_local_euler_projection {p : ℕ} (hp : p.Prime)
  (dSelected : Bool) :
  localProjection p dSelected =
    if dSelected then -(1 : ℝ) / (p : ℝ)
    else 1 - 1 / ((p : ℝ) * ((p : ℝ) - 1))
```

A.3.2 GoldbachActualDivisorFibreBound

Namespace: `GoldbachResearch.ActualDivisorFibreBound`.

The actual finite cutoff is retained. This is the coefficient bound used by a later written weighted-distribution argument, not that distribution theorem.

```
theorem actual_truncated_fibre_abs_bound {r s alpha : ℕ}
  (hr : 1 ≤ r) (hs : 1 ≤ s) (ha : 1 ≤ alpha) :
  |signedDivisorCoefficient r s alpha| ≤
    (s.divisors.card : ℝ) * Real.log (alpha : ℝ)
```

A.3.3 GoldbachActualTerminalCutoffInvariant

Namespace: `GoldbachResearch.ActualTerminalCutoffInvariant`.

These identities retain the actual kernel and genuine Vaughan coefficients. Changing the U cut transfers terms between the two rows; it does not make their sum small or discharge a covered-support hypothesis.

```
theorem actual_rows_complete (N alpha Q U V : ℕ) :
  fullRow N alpha Q U V + smallRow N alpha Q U V =
  primeRow N alpha Q V
```

```
theorem actual_terminal_defect (N alpha Q U V : ℕ) (e : ℝ) :
  ledgerDeficit N alpha Q U V e - smallRow N alpha Q U V =
  -primeRow N alpha Q V + e + |e|
```

```

theorem actual_terminal_U_invariant (N alpha Q U U' V : ℕ) (e e' : ℝ) :
  (ledgerDeficit N alpha Q U V e - smallRow N alpha Q U V) -
  (ledgerDeficit N alpha Q U' V e' - smallRow N alpha Q U' V) =
  (e + |e|) - (e' + |e'|)

```

A.3.4 GoldbachCompleteHPrimeOrbit

Namespace: GoldbachResearch.CompleteHPrimeOrbit.

The complete arithmetic coefficient is retained. The last inequality allows arbitrary real high, requires nonnegative low, and exposes the boundary through $\max(\text{high}-\text{low}, 0)$; it does not assert favorable closure of the actual selectors.

```

theorem H_literal_finite_sum (y n : ℕ) :
  H y n = ∑ d ∈ n.divisors, ∑ uv ∈ d.divisorsAntidiagonal,
    (if y < uv.1 then ArithmeticFunction.moebius uv.1 else 0) *
    (if y < uv.2 then ArithmeticFunction.moebius uv.2 else 0)

theorem H_rough_formula {y n : ℕ} (hy : 1 ≤ y) (hn : 0 < n)
  (hr : rough y n) :
  H y n = ArithmeticFunction.moebius n + 1 - 2 * (if n = 1 then 1 else 0)

theorem prime_divisor_orbit_partition {p L k : ℕ} (hp : p.Prime)
  (hk : k | p * L) :
  (k | L ∧ ¬p | k) ∨ ∃ x, x | L ∧ k = p * x

theorem H_one_small_prime {y p e : ℕ} (hy : 1 ≤ y) (hp : p.Prime)
  (hpy : p ≤ y) (he : 0 < e) (hr : rough y e) :
  H y (p * e) = -H y e

theorem actual_complete_orbit_upper {y p a e b : ℕ}
  (hy : 1 ≤ y) (hp : p.Prime) (hpy : p ≤ y)
  (ha : 0 < a) (hra : rough y a) (he : 0 < e) (hre : rough y e)
  (hb : 1 ≤ b) (low high : ℝ) (hlow : 0 ≤ low) :
  0 ≤ (H y a : ℝ) * Real.log b * (H y e : ℝ) * low * Real.log p ∧
  (H y a : ℝ) * Real.log b *
    ((H y e : ℝ) * high * Real.log e +
     (H y (p*e) : ℝ) * low * Real.log (p*e))
  ≤ (H y a : ℝ) * Real.log b * (H y e : ℝ) *
    (max (high-low) 0 * Real.log e - low * Real.log p)

```

A.3.5 GoldbachLiteralDivisorCompletion

Namespace: GoldbachResearch.LiteralDivisorCompletion.

These conclusions preserve the integer floor cap and strict face. They require the written proper-divisor or prime-roughness assumptions; no analytic sign or size estimate for W is included.

```

theorem full_literal_divisor_sum (m : ℕ) (hm : m ≠ 0) :
  (∑ k ∈ m.divisors, muReal k * Real.log ((k : ℝ) / (m : ℝ))) =
    -ArithmeticFunction.vonMangoldt m

theorem literal_completion_of_proper_admission (alpha Q m : ℕ) (hm : m ≠ 0)
  (hadmit : ∀ k ∈ m.divisors, k ≠ m → k ≤ Q ∧ alpha * k < m) :
  divisorKernel alpha Q m = -ArithmeticFunction.vonMangoldt m

theorem actual_floor_cap_admits_proper_divisors (N alpha m : ℕ)
  (hm : m ≠ 0) (hN : m ≤ N - 1) (ha : 0 < alpha)
  (hcofactor : ∀ c ∈ m.divisors, c ≠ 1 → alpha < c) :
  ∀ k ∈ m.divisors, k ≠ m →
    k ≤ (N - 1) / alpha ∧ alpha * k < m

theorem literal_completion_at_original_floor (N alpha m : ℕ)
  (hm : m ≠ 0) (hN : m ≤ N - 1) (ha : 0 < alpha)
  (hcofactor : ∀ c ∈ m.divisors, c ≠ 1 → alpha < c) :
  divisorKernel alpha ((N - 1) / alpha) m =
    -ArithmeticFunction.vonMangoldt m

theorem rough_prime_factors_control_cofactors (alpha m : ℕ) (hm : m ≠ 0)
  (hrough : ∀ p : ℕ, Nat.Prime p → p | m → alpha < p) :
  ∀ c ∈ m.divisors, c ≠ 1 → alpha < c

theorem rough_literal_completion_at_original_floor (N alpha m : ℕ)
  (hm : m ≠ 0) (hN : m ≤ N - 1) (ha : 0 < alpha)
  (hrough : ∀ p : ℕ, Nat.Prime p → p | m → alpha < p) :
  divisorKernel alpha ((N - 1) / alpha) m =
    -ArithmeticFunction.vonMangoldt m

```

A.3.6 GoldbachActualSmallDivisorCredit

Namespace: `GoldbachResearch.ActualSmallDivisorCredit`.

The support equality is proved from roughness of L and the full proper-divisor condition on P . The negative coefficient does not by itself prove that a sector contains enough prime first-axis mass.

```

theorem literal_small_divisors_product {alpha P L : ℕ}
  (hP : alpha < P) (hL : 0 < L)
  (hproper : ∀ d ∈ P.divisors, d ≠ P → d ≤ alpha)
  (hrough : ∀ p : ℕ, p.Prime → p | L → alpha < p) :
  (P * L).divisors.filter (fun d => d ≤ alpha) = P.divisors.erase P

theorem actual_small_divisor_credit {alpha P L : ℕ}
  (hP : alpha < P) (hL : 0 < L)
  (hproper : ∀ d ∈ P.divisors, d ≠ P → d ≤ alpha)
  (hrough : ∀ p : ℕ, p.Prime → p | L → alpha < p) :
  smallDivisorCredit alpha (P * L) = ArithmeticFunction.vonMangoldt P +
    (ArithmeticFunction.moebius P : ℝ) * Real.log (P : ℝ)

theorem actual_negative_composite_credit {alpha P L : ℕ}
  (hP : alpha < P) (hL : 0 < L)
  (hproper : ∀ d ∈ P.divisors, d ≠ P → d ≤ alpha)
  (hrough : ∀ p : ℕ, p.Prime → p | L → alpha < p)
  (hpow : ¬IsPrimePow P) (hmu : ArithmeticFunction.moebius P = -1) :
  smallDivisorCredit alpha (P * L) = -Real.log (P : ℝ)

```

A.4 Source hashes and reproduction references

The exact source identifiers are as follows. Full production and independent-replay receipt paths and hashes are in `lean_inventory.json`; source-copy equality, compiler exit zero, and log/compiled-artifact hashes were checked against those receipts. Compiled-artifact hashes can differ between fresh directories because build paths are embedded; each must match its own receipt.

`GoldbachActualEulerProjection.lean`

SHA256 7521a244da3f6000d4841156dee78f819bf9a1e075d6da127dacf6c0170520c0

`GoldbachActualDivisorFibreBound.lean`

SHA256 f9cdd1d266889b5f672ef76c35b845e51e1b81a6d3bedcd49737b01c26281e44

`GoldbachActualTerminalCutoffInvariant.lean`

SHA256 52a6a5b0d0f59879d2567df1415d8d4d9e2f5ff695a57421db18282a97a5c039

`GoldbachCompleteHPrimeOrbit.lean`

SHA256 df14ac670788ddbf2091cc62c41b93156267425ef937927f12945edf5f2ffa5e

`GoldbachLiteralDivisorCompletion.lean`

SHA256 fd3c4885ff79241c25b977af1954cbd8c1512e133dc4ccf9f391b6558c474fde

`GoldbachActualSmallDivisorCredit.lean`

SHA256 25feafb7ce3ce408ec76cd83e36682e947b508d0e580cd419ab6630c62c11a16

The read-only reproduction folder contains the 71 canonical sources and their dependency-closed local import graph, the original compile scripts, retained source-bound receipts and axiom logs, and a byte-hash manifest. Its original scripts record the authoring host's absolute paths. They are evidence and templates, not automatically portable commands; a replay on another host must resolve equivalent pinned libraries and build into a new empty directory.

The original accepted fresh replays imported exactly the eight dependency library roots aesop, batteries, importGraph, LeanSearchClient, mathlib, plausible, proofwidgets and Qq. Standard Lean system libraries are implicit. The reproduction folder contains no dependency cache or copied olean that could silently replace a requested fresh compilation.

The immutable latest-cycle archives are identified by:

```
base snapshot (16 new public conclusions):
3b88e4f9367fb2b9718450df362149d1f4bcfa1fda886851e0db22091c35d720
continuation addendum (3 new public conclusions; combined19):
66dc5eecb878c78a260a469200bb6f81d48d8b14146b5fd4a06a357c7b1a5347
post-freeze rough-sector supplement (0 new Lean conclusions):
9d44c84e76e78af0a0de3219881a821976f397688c5db271fa21b6034eff1442
```

Independent exterior receipts checked every CRC, hash, payload size and byte correspondence, and the unchanged prior archives. These integrity checks establish provenance, not another mathematical estimate.

No declaration above formalizes Davenport, a large-sieve or Bombieri–Vinogradov input, a zero-avoiding contour, or the lower-sieve density used by the written alpha-eighth credit theorem. The negative aggregate credit has a separate qualitative, unevaluated/ineffective onset and a still uncontrolled complement. A one-sided transfer to a positive prime-pair mass also requires the stated sufficient reference and loss reserve. Neither the 19 latest finite certificates nor the 658-entry historical/current inventory discharges the remaining signed arithmetic estimate or proves Goldbach.

References

- [1] G. H. Hardy and J. E. Littlewood, *Some problems of Partitio Numerorum; III: On the expression of a number as a sum of primes*, Acta Mathematica 44 (1923), 1–70. <https://doi.org/10.1007/BF02403921>.
- [2] G. Bhowmik and L. Grimmelt, *The exceptional set of the Goldbach problem*, arXiv:2607.27282v2, 13 August 2026. <https://arxiv.org/abs/2607.27282v2>.
- [3] T. Oliveira e Silva, S. Herzog and S. Pardi, *Empirical verification of the even Goldbach conjecture and computation of prime gaps up to $4 \cdot 10^{18}$* , Mathematics of Computation 83 (2014), 2033–2060. <https://doi.org/10.1090/S0025-5718-2013-02787-1>.
- [4] J.-J. Huang and H. Li, *On the connection between the Goldbach conjecture and the Elliott–Halberstam conjecture*, arXiv:2005.03811v2, 28 August 2022. <https://arxiv.org/abs/2005.03811v2>.
- [5] H. L. Montgomery and R. C. Vaughan, *The large sieve*, Mathematika 20 (1973), 119–134. <https://doi.org/10.1112/S0025579300004708>.
- [6] D. A. Goldston and C. Y. Yildirim, *Higher correlations of divisor sums related to primes II: variations of the error term in the prime number theorem*, Proceedings of the London Mathematical Society 95 (2007), 199–247. <https://arxiv.org/abs/math/0412366>.
- [7] A. Akbary and K. Hambrook, *A variant of the Bombieri–Vinogradov theorem with explicit constants and applications*, arXiv:1309.2730v2. Author-hosted final text: <https://www.cs.uleth.ca/~akbary/EBVT-Final.pdf>.
- [8] M. Bordignon, *Medium-sized values for the prime number theorem for primes in arithmetic progressions*, New York Journal of Mathematics 27 (2021), 1415–1438. <https://nyjm.albany.edu/j/2021/27-54p.pdf>.
- [9] D. R. Johnston and A. Yang, *Some explicit estimates for the error term in the prime number theorem*, arXiv:2204.01980v2. <https://arxiv.org/abs/2204.01980v2>.
- [10] E. S. Lee and N. Leong, *New explicit bounds for Mertens function and the reciprocal of the Riemann zeta-function*, arXiv:2208.06141v5, 9 September 2026. <https://arxiv.org/abs/2208.06141v5>. The version is material to the constants used here.
- [11] M. A. Bennett, G. Martin, K. O’Bryant and A. Rechnitzer, *Explicit bounds for primes in arithmetic progressions*, Illinois Journal of Mathematics 62 (2018), 427–532. <https://personal.math.ubc.ca/~bennett/BeMaObRe-IJM-2018.pdf>.
- [12] A. Sedunova, *A logarithmic improvement in the Bombieri–Vinogradov theorem*, Journal de Théorie des Nombres de Bordeaux 31 (2019), 635–651. <https://jtnb.centre-mersenne.org/articles/10.5802/jtnb.1098/>.
- [13] K. Matomäki and S. Zuniga–Altermann, *Weighted sieves with switching*, Mathematical Proceedings of the Cambridge Philosophical Society 179 (2025), 351–372. <https://doi.org/10.1017/S0305004125000258>.
- [14] H.-E. Richert, *Lectures on Sieve Methods*, Tata Institute of Fundamental Research, Bombay (1976). <https://mathweb.tifr.res.in/Documents/Publications/Lectures/tifr55.pdf>.

- [15] J. B. Rosser and L. Schoenfeld, *Approximate formulas for some functions of prime numbers*, Illinois Journal of Mathematics 6 (1962), 64–94. <https://doi.org/10.1215/ijm/1255631807>.
- [16] Y. Motohashi, *Lectures on Sieve Methods and Prime Number Theory*, Tata Institute of Fundamental Research, Bombay (1983). Accessible text: https://www.math.utoledo.edu/~codenth/Spring_13/3200/NT-books/Lectures_on_Sieve_Methods_and_Prime_Number_Theory-Motohashi.pdf.
- [17] S. Durand, *goldbach-horizon*, Project Horizon public repository, commit 979c05416140504f11c6e9ba6ad1dd565838bea1. <https://github.com/seisner1967-hash/goldbach-horizon>. Source and interface audit retrieved 1 October 2026.
- [18] S. Durand, *CS21: The Shadow Equation*, Project Horizon computational note, version9.0 (January 2026), supplied corpus. <https://drive.google.com/file/d/1KxZSfj0eWZ7zfwYhKgovQbaWzdysnwLY/view>.
- [19] L. de Moura and S. Ullrich, *The Lean4 theorem prover and programming language*, CADE28 (2021), 625–635. https://doi.org/10.1007/978-3-030-79876-5_37.
- [20] The Mathlib Community, *Mathlib4*, mathematical library, revision 9837ca9d65d9de6fad1ef4381750ca688774e608. <https://github.com/leanprover-community/mathlib4>.
- [21] arXiv, *Moderation: content guidelines and policy on generative AI*. <https://info.arxiv.org/help/moderation/index.html>. Accessed 1 October 2026.